



UNIVERSIDADE D  
COIMBRA

Vasco Tomás Ramos Rodrigues

A CLINICIAN-VALIDATED LLM PIPELINE FOR  
STRUCTURING ONCOLOGY DATA AND  
CLINICAL TRIAL ELIGIBILITY MATCHING

Dissertation in the context of the Master in Informatics Engineering, Artificial Intelligence, advised by Professor Pedro Furtado and presented to the Department of Informatics Engineering of the Faculty of Sciences and Technology of the University of Coimbra.

January 2026





DEPARTAMENTO DE  
ENGENHARIA INFORMÁTICA  
**FACULDADE DE  
CIÊNCIAS E TECNOLOGIA  
UNIVERSIDADE D  
COIMBRA**

Vasco Tomás Ramos Rodrigues

# **A CLINICIAN-VALIDATED LLM PIPELINE FOR STRUCTURING ONCOLOGY DATA AND CLINICAL TRIAL ELIGIBILITY MATCHING**

**Dissertation in the context of the Master in Informatics Engineering,  
specialization in Artificial Intelligence, advised by Professor Pedro Furtado  
and presented to the Department of Informatics Engineering of the Faculty of  
Sciences and Technology of the University of Coimbra.**

January 2026





DEPARTAMENTO DE  
ENGENHARIA INFORMÁTICA

FACULDADE DE  
CIÊNCIAS E TECNOLOGIA  
UNIVERSIDADE DE  
COIMBRA

Vasco Tomás Ramos Rodrigues

**UMA PIPELINE DE LLM VALIDADA POR  
CLÍNICOS PARA ESTRUTURAR DADOS  
ONCOLÓGICOS E AVALIAR A  
ELEGIBILIDADE PARA ENSAIOS CLÍNICOS**

Dissertação no âmbito do Mestrado em Engenharia Informática,  
especialização em Inteligência Artificial, orientada pelo Professor Pedro  
Furtado e apresentada ao Departamento de Engenharia Informática da  
Faculdade de Ciências e Tecnologia da Universidade de Coimbra.

janeiro 2026



## STATEMENT ON THE USE OF ARTIFICIAL INTELLIGENCE TOOLS

This document was reviewed and refined with the assistance of Large Language Models, such as ChatGPT, or similar tools, which helped check grammar, correct typos, and enhance clarity. Any Artificial Intelligence-generated text or content is clearly indicated within the document and is used only to a limited extent. The overall content and ideas remain solely the responsibility of the author.



# Abstract

Oncology departments generate large volumes of unstructured clinical data through free-text diaries and medical notes, which poses significant challenges for information management, clinical decision support, and research activities. One of the most critical consequences of this lack of structure is the persistent under-enrollment of eligible patients in clinical trials, as matching patient profiles with complex inclusion and exclusion criteria remains a largely manual, time-consuming, and error-prone process.

This dissertation proposes a system that leverages Large Language Models (LLMs) to transform unstructured oncology data into structured, validated patient profiles and to support the identification of eligible patients for clinical trials. The system operates in two main phases. First, an LLM-based extraction pipeline processes free-text clinical diaries to identify and normalize hyper-relevant clinical parameters, which are subsequently validated by clinicians and stored in a relational database. Second, clinical trial documentation is processed to extract inclusion and exclusion criteria, which are translated into machine-readable logical rules and validated prior to use. These validated criteria are then deterministically evaluated against the structured patient database to ensure transparent and reproducible eligibility decisions.

In addition to patient-trial matching, the system supports monitoring the quality of care by computing survival metrics such as Overall Survival (OS) and Progression-Free Survival (PFS), applying Kaplan–Meier survival analysis, and providing interactive visualizations for clinical assessment. By combining LLM-based information extraction with clinician validation and deterministic evaluation, the proposed approach reduces administrative burden, improves traceability, and enhances efficiency in clinical trial enrolment and outcome analysis, contributing to a more data-driven oncology workflow.

## Keywords

Large Language Models; Information Extraction; Clinical Trial Eligibility; Unstructured Clinical Data; Clinician-in-the-Loop; Deterministic Eligibility Evaluation; Survival Analysis; Structured Patient Data



## Resumo

Os serviços de oncologia geram grandes volumes de dados clínicos não estruturados através de diários clínicos e notas médicas em texto livre, o que representa desafios significativos para a gestão de informação, apoio à decisão clínica e atividades de investigação. Uma das consequências mais críticas desta falta de estrutura é a subinscrição persistente de doentes elegíveis em ensaios clínicos, uma vez que o processo de correspondência entre perfis de pacientes e critérios complexos de inclusão e exclusão continua a ser maioritariamente manual, moroso e propenso a erros.

Esta dissertação propõe um sistema que recorre a Large Language Models (LLMs) para transformar dados oncológicos não estruturados em perfis de pacientes estruturados e validados, apoiando simultaneamente a identificação de doentes elegíveis para ensaios clínicos. O sistema opera em duas fases principais. Primeiro, uma pipeline de extração baseada em LLM processa diários clínicos em texto livre para identificar e normalizar parâmetros clínicos hiper-relevantes, que são posteriormente validados por clínicos e armazenados numa base de dados relacional. Em seguida, a documentação dos ensaios clínicos é processada para extrair critérios de inclusão e exclusão, que são traduzidos em regras lógicas legíveis por máquina e validadas antes da sua utilização. Estes critérios validados são então avaliados de forma determinística sobre a base de dados estruturada de pacientes, garantindo decisões de elegibilidade transparentes e reproduzíveis.

Para além da correspondência entre pacientes e ensaios, o sistema suporta a monitorização da qualidade dos cuidados através do cálculo de métricas de sobrevivência, como Overall Survival (OS) e Progression-Free Survival (PFS), aplicando análise de sobrevivência de Kaplan–Meier e fornecendo visualizações interativas para avaliação clínica. Ao combinar extração de informação baseada em LLM com validação clínica e avaliação determinística, a abordagem proposta reduz a carga administrativa, melhora a rastreabilidade e aumenta a eficiência no processo de recrutamento para ensaios clínicos e na análise de resultados, contribuindo para um fluxo de trabalho oncológico mais orientado por dados.

## Palavras-Chave

Modelos de Linguagem de Grande Escala; Extração de Informação; Elegibilidade para Ensaios Clínicos; Dados Clínicos Não Estruturados; Clínico-no-Ciclo; Avaliação Determinística de Elegibilidade; Análise de Sobrevivência; Dados Estruturados de Pacientes



# Contents

|          |                                                                        |           |
|----------|------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                                    | <b>1</b>  |
| 1.1      | Motivation . . . . .                                                   | 1         |
| 1.2      | As-Is Flow . . . . .                                                   | 2         |
| 1.3      | To-Be Flow . . . . .                                                   | 3         |
| 1.4      | Contributions . . . . .                                                | 4         |
| 1.5      | Structure . . . . .                                                    | 5         |
| <b>2</b> | <b>Methodology and Project Planning</b>                                | <b>6</b>  |
| 2.1      | Methodology . . . . .                                                  | 6         |
| 2.1.1    | Sprint structure and ceremonies . . . . .                              | 6         |
| 2.1.2    | Management Tools . . . . .                                             | 7         |
| 2.2      | Work Conducted in the First Semester . . . . .                         | 7         |
| 2.3      | Work Plan for the Second Semester . . . . .                            | 8         |
| 2.4      | Risk Management . . . . .                                              | 9         |
| <b>3</b> | <b>Related Work</b>                                                    | <b>12</b> |
| 3.1      | Extraction of Unstructured Clinical Information through LLMs . . . . . | 12        |
| 3.1.1    | Evaluating LLM Training and Prompting Strategies . . . . .             | 13        |
| 3.1.2    | Comparative Evaluation of LLMs for Information Extraction . . . . .    | 13        |
| 3.1.3    | Efficiency of LLMs Compared to Manual Review . . . . .                 | 13        |
| 3.1.4    | Large-Scale Model Benchmarking for Data Extraction . . . . .           | 14        |
| 3.1.5    | Extraction of Social Determinants of Health (SDoH) . . . . .           | 14        |
| 3.1.6    | Oncology Information Extraction with Expert-Annotated Data . . . . .   | 15        |
| 3.1.7    | Summary . . . . .                                                      | 15        |
| 3.2      | Matching Patients for Clinical Trials with LLMs . . . . .              | 15        |
| 3.2.1    | Zero-Shot Matching and the TrialGPT Framework . . . . .                | 15        |
| 3.2.2    | Retrieval Pipelines and LLM-Based Re-Ranking . . . . .                 | 16        |
| 3.2.3    | Ontology-Enhanced Prompt-Baed Matching . . . . .                       | 17        |
| 3.2.4    | Distilling Open-Source Models for Patient-Trial Matching . . . . .     | 17        |
| 3.2.5    | End-to-End Matching with Real-World Validation . . . . .               | 18        |
| 3.2.6    | Scalable Zero-Shot Matching Systems . . . . .                          | 18        |
| 3.2.7    | Summary . . . . .                                                      | 19        |
| 3.3      | Conclusion . . . . .                                                   | 19        |
| <b>4</b> | <b>Approach</b>                                                        | <b>20</b> |
| 4.1      | Method Overview . . . . .                                              | 20        |
| 4.2      | Hyper-Relevant Parameter Extraction . . . . .                          | 21        |
| 4.3      | Application of Clinical Trial Criteria . . . . .                       | 21        |
| 4.4      | Clinical Documents generalization . . . . .                            | 22        |

---

|          |                                                                                    |           |
|----------|------------------------------------------------------------------------------------|-----------|
| 4.5      | Choices and Tools . . . . .                                                        | 24        |
| 4.5.1    | Usage of Language Models . . . . .                                                 | 24        |
| 4.5.2    | JSON structure and PostgreSQL . . . . .                                            | 25        |
| 4.5.3    | Conversion of Clinical Criteria and Deterministic evaluation of patients . . . . . | 25        |
| 4.5.4    | Pipeline . . . . .                                                                 | 26        |
| <b>5</b> | <b>Requirement Analysis and Theme</b>                                              | <b>27</b> |
| 5.1      | Context and Stakeholder Needs . . . . .                                            | 27        |
| 5.2      | Automated Structured Data Extraction . . . . .                                     | 28        |
| 5.3      | Identification of Eligible Patients for Clinical Trials . . . . .                  | 30        |
| 5.4      | Monitoring the quality of care . . . . .                                           | 31        |
| <b>6</b> | <b>Architecture</b>                                                                | <b>33</b> |
| 6.1      | Context . . . . .                                                                  | 33        |
| 6.2      | Container . . . . .                                                                | 34        |
| 6.3      | Component . . . . .                                                                | 35        |
| <b>7</b> | <b>Experimental Setup</b>                                                          | <b>37</b> |
| 7.1      | Experimental Objectives . . . . .                                                  | 37        |
| 7.2      | Dataset and Data Preparation . . . . .                                             | 37        |
| 7.3      | LLMs Evaluated . . . . .                                                           | 38        |
| 7.4      | Prompting and Inference Configuration . . . . .                                    | 38        |
| 7.5      | Evaluation Metrics . . . . .                                                       | 39        |
| 7.6      | Experimental Scenarios . . . . .                                                   | 39        |
| <b>8</b> | <b>Conclusion</b>                                                                  | <b>40</b> |
|          | <b>References</b>                                                                  | <b>41</b> |
|          | <b>Appendix A Requirements</b>                                                     | <b>46</b> |
| A.1      | Sea Level Requirements . . . . .                                                   | 47        |
| A.1.1    | System Requirements . . . . .                                                      | 47        |
| A.1.2    | Identification of Eligible Patients for Clinical Trials . . . . .                  | 48        |
| A.1.3    | Monitoring the Quality of Care . . . . .                                           | 50        |
| A.1.4    | Automated Structured Data Extraction . . . . .                                     | 52        |
| A.2      | Oyster Level Requirements . . . . .                                                | 53        |
| A.2.1    | FR0.1 - Load historical data . . . . .                                             | 53        |
| A.2.2    | FR0.2 - Pre-process clinical documents . . . . .                                   | 53        |
| A.2.3    | FR1.1 - Extract clinical trial criteria . . . . .                                  | 54        |
| A.2.4    | FR1.2 - Convert Criteria into logic . . . . .                                      | 54        |
| A.2.5    | FR1.3 - Validation of inclusion and exclusion criteria . . . . .                   | 54        |
| A.2.6    | FR1.4 - Compare criteria with database . . . . .                                   | 55        |
| A.2.7    | FR1.5 - Generate eligibility report . . . . .                                      | 55        |
| A.2.8    | FR1.6 - Saving included patients . . . . .                                         | 56        |
| A.2.9    | FR2.1 - Retrieve treatment data . . . . .                                          | 56        |
| A.2.10   | FR2.2 - Compute OS and PFS . . . . .                                               | 57        |
| A.2.11   | FR2.3 - Apply Kaplan-Meier analysis . . . . .                                      | 57        |
| A.2.12   | FR2.4 - Visualize results . . . . .                                                | 58        |

|                   |                                                                      |           |
|-------------------|----------------------------------------------------------------------|-----------|
| A.2.13            | FR3.1 - Select Diaries for extraction . . . . .                      | 59        |
| A.2.14            | FR3.2 - Extract structured info . . . . .                            | 59        |
| A.2.15            | FR3.3 - Normalize text . . . . .                                     | 61        |
| A.2.16            | FR3.4 - Validation of extracted parameters . . . . .                 | 63        |
| A.2.17            | FR3.5 - Store structured profiles . . . . .                          | 63        |
| <b>Appendix B</b> | <b>Abstract Templates</b>                                            | <b>65</b> |
| B.0.1             | Clinical Trial Template . . . . .                                    | 65        |
| B.0.2             | Clinical Diary Template . . . . .                                    | 66        |
| <b>Appendix C</b> | <b>Background Knowledge</b>                                          | <b>68</b> |
| C.1               | Symbolic and Connectionist AI . . . . .                              | 68        |
| C.2               | Language Models . . . . .                                            | 69        |
| C.2.1             | From Simple Models to Probabilistic Language Understanding . . . . . | 69        |
| C.2.2             | Examples of Prediction Tasks . . . . .                               | 70        |
| C.2.3             | Evaluating Language Models . . . . .                                 | 71        |
| C.2.4             | Training Process . . . . .                                           | 71        |
| C.3               | Simpler Generative Language Models (N-gram) . . . . .                | 71        |
| C.3.1             | Co-occurrence Matrix . . . . .                                       | 72        |
| C.3.2             | Predicting the Next Word . . . . .                                   | 72        |
| C.3.3             | Semantic Properties of Word Embeddings . . . . .                     | 73        |
| C.3.4             | Generalized Autoregressive L-gram Model . . . . .                    | 73        |
| C.4               | Recipe for a Large Language Model . . . . .                          | 74        |
| C.4.1             | Feedforward Layers . . . . .                                         | 74        |
| C.4.2             | Attention Mechanism . . . . .                                        | 74        |
| C.4.3             | Attention Heads . . . . .                                            | 75        |
| C.4.4             | Positional Encodings . . . . .                                       | 76        |
| C.5               | Attention Is All You Need . . . . .                                  | 76        |
| C.5.1             | Model Architecture . . . . .                                         | 76        |
| C.5.2             | Encoder . . . . .                                                    | 77        |
| C.5.3             | Decoder . . . . .                                                    | 77        |
| C.5.4             | The Attention Mechanism . . . . .                                    | 77        |
| C.5.5             | Scaled Dot-Product Attention . . . . .                               | 78        |
| C.5.6             | Applications of Attention in the Transformer . . . . .               | 78        |
| C.6               | Prompting Strategies . . . . .                                       | 79        |
| C.7               | Information Extraction . . . . .                                     | 80        |
| <b>Appendix D</b> | <b>Mockups</b>                                                       | <b>82</b> |





# Acronyms

**AI** Artificial Intelligence.

**DB** DataBase.

**DR** Data Requirement.

**ECOG-PS** Eastern Cooperative Oncology Group Performance Status.

**FFN** Feedforward Neural Network.

**FR** Functional Requirement.

**IE** Information Extraction.

**IR** Interface Requirement.

**JSON** JavaScript Object Notation.

**KM** Kaplan–Meier.

**LLM** Large Language Models.

**ML** Machine Learning.

**NER** Named Entity Recognition.

**OS** Overall Survival.

**PCA** Principal Component Analysis.

**PDF** Portable Document Format.

**PFS** Progression-Free Survival.

**RNN** Recurrent Neural Network.

**SDoH** Social Determinants of Health.

**SPSS** Statistical Package for the Social Sciences.

**UI** User Interface.

**ULS** Unidade de Saúde Local.

# List of Figures

|     |                                                  |    |
|-----|--------------------------------------------------|----|
| 1.1 | Current workflow for patient screening . . . . . | 3  |
| 1.2 | Proposed solution workflow . . . . .             | 4  |
| 2.1 | Gantt Diagram of first semester . . . . .        | 8  |
| 2.2 | Gantt Diagram of second semester . . . . .       | 9  |
| 4.1 | Clinical Trial Abstraction . . . . .             | 23 |
| 4.2 | Clinical Diary Abstraction . . . . .             | 24 |
| 5.1 | Parameter Extraction Workflow . . . . .          | 29 |
| 5.2 | Patient-Trial Matching workflow . . . . .        | 31 |
| 6.1 | Context Diagram of the system . . . . .          | 34 |
| 6.2 | Container Diagram of the system . . . . .        | 35 |
| 6.3 | Component Diagram of the system . . . . .        | 36 |
| D.1 | Clinical diary field extraction page . . . . .   | 83 |
| D.2 | Clinical diary upload page . . . . .             | 83 |
| D.3 | Clinical trial detail page . . . . .             | 84 |
| D.4 | Elected patients page . . . . .                  | 84 |
| D.5 | Import clinical trial page . . . . .             | 85 |
| D.6 | Main-page . . . . .                              | 85 |
| D.7 | Survivability analysis page . . . . .            | 86 |



# List of Tables

|      |                                                                                                  |    |
|------|--------------------------------------------------------------------------------------------------|----|
| 2.1  | Risk Assessment Metrics and Classification Criteria . . . . .                                    | 10 |
| 2.2  | Project Risks and Mitigation Strategies . . . . .                                                | 11 |
| 5.1  | Automated Structured Data Extraction - Functional Requirements .                                 | 30 |
| 5.2  | Identification of Eligible Patients for Clinical Trials – Functional Re-<br>quirements . . . . . | 31 |
| 5.3  | Monitoring the Quality of Care – Functional Requirements . . . . .                               | 32 |
| A.1  | System Requirements - Functional Requirements . . . . .                                          | 47 |
| A.2  | Identification of Eligible Patients for Clinical Trials – Functional Re-<br>quirements . . . . . | 48 |
| A.3  | Identification of Eligible Patients for Clinical Trials - Data Require-<br>ments . . . . .       | 48 |
| A.4  | Identification of Eligible Patients for Clinical Trials - Interface Re-<br>quirements . . . . .  | 49 |
| A.5  | Monitoring the Quality of Care – Functional Requirements . . . . .                               | 50 |
| A.6  | Mapping of Extracted Fields to Necessary Fields . . . . .                                        | 51 |
| A.7  | Monitoring the Quality of Care - Data Requirements . . . . .                                     | 51 |
| A.8  | Monitoring the Quality of Care - Interface Requirements . . . . .                                | 51 |
| A.9  | Automated Structured Data Extraction - Functional Requirements .                                 | 52 |
| A.10 | Automated Structured Data Extraction - Data Requirements . . . . .                               | 53 |
| A.11 | Mapping of Extracted Fields to Necessary Fields . . . . .                                        | 57 |
| C.1  | 3-gram Co-occurrence Matrix . . . . .                                                            | 72 |
| C.2  | Decoder Time Steps and Target Outputs . . . . .                                                  | 77 |



# Chapter 1

## Introduction

This dissertation reports the development of a clinical decision-support system carried out in collaboration with the Unidade Local de Saúde (ULS) de Coimbra, specifically within the oncology department. The project was conducted in a real-world hospital context, addressing practical challenges faced by oncologists in their daily clinical and research activities. Its primary objective is to support the structuring of unstructured patient clinical data and to enable its systematic comparison with clinical trial eligibility criteria, with the aim of improving efficiency in patient screening and trial enrolment processes.

The work was developed in close collaboration with an oncology specialist, Dra. Judy, who acted as the main clinical stakeholder. Her expertise and continuous feedback were fundamental in shaping the system requirements, validating the clinical relevance of the extracted data, and ensuring alignment with real clinical workflows. The target users of the proposed system are oncology clinicians involved in patient follow-up and clinical research, particularly those responsible for identifying and enrolling eligible patients into clinical trials.

By being grounded in an operational oncology department and informed by direct clinical input, this project seeks to bridge the gap between advanced Artificial Intelligence techniques and their practical applicability in hospital environments, ensuring that the proposed solution remains clinically meaningful, usable, and ethically aligned with existing medical decision-making processes.

### 1.1 Motivation

Oncology departments handle a vast number of patients on a daily basis, generating large volumes of complex and unstructured clinical data. This reality, common across many healthcare systems, is characterized by the extensive use of free-text clinical notes, resulting in a highly unstandardized set of medical records. This poses significant challenges for information management, as critical data required for patient monitoring and research support remains embedded in formats that are difficult to process and analyze.

Oncologists, who already operate under high cognitive and emotional demands, face a substantial administrative burden due to the manual extraction and interpretation of this information. This contributes to operational inefficiency and clinician burnout. One of the most critical consequences of this scenario is the persistent under-enrollment of eligible patients in clinical trials. When patient information recorded in clinical diaries cannot be effectively matched with structured trial eligibility criteria, institutions risk financial penalties and miss opportunities to advance clinical research.

This project aims to address these challenges through the use of Large Language Models (LLMs), which offer a promising approach to processing unstructured textual data. Unlike traditional solutions that rely exclusively on structured inputs, LLMs are capable of interpreting free-text clinical narratives, despite variations in writing style and the presence of inconsistencies. By leveraging LLMs to support the extraction, normalization, and analysis of unstructured oncology data, this project seeks to assist clinicians in the identification of patients eligible for clinical trials. The ultimate goals are to reduce the burden of manual patient screening and to contribute to a more efficient and data-driven oncology research workflow.

## 1.2 As-Is Flow

Currently, the process of identifying eligible patients for clinical trials is highly repetitive and predominantly manual. Clinicians are required to compare each patient against multiple clinical trials, each containing extensive inclusion and exclusion criteria. This task is often performed alongside routine clinical activities and is frequently interrupted by consultations and other duties. Such interruptions force clinicians to repeatedly revisit trial documentation, further reducing efficiency.

The lack of standardized clinical records significantly exacerbates this problem. As a result, clinicians spend a considerable amount of time performing administrative tasks that could otherwise be allocated to direct patient care or research-related activities.

Given this context, there is a clear need to standardize patient clinical data and to support the automated comparison between patient profiles and clinical trial eligibility criteria.

All that was described can be seen on the Figure 1.1.

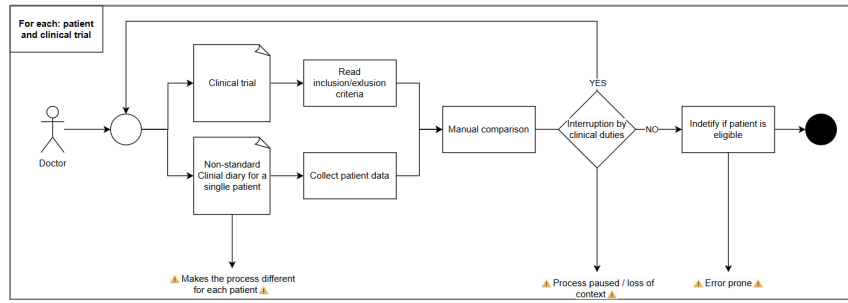


Figure 1.1: Current workflow for patient screening

### 1.3 To-Be Flow

The proposed solution aims to address the limitations identified in the current workflow by introducing a structured, semi-automated process supported by Large Language Models and clinician validation. The system assists clinicians by transforming unstructured clinical diaries into structured patient profiles through an LLM-based extraction pipeline. These extracted parameters are reviewed and validated by the clinician before being stored in a centralized database, eliminating the need for repeated manual data interpretation in future analyses.

In parallel, clinical trial documentation is processed to extract inclusion and exclusion criteria, which are subsequently translated into structured logical rules. These rules are also subject to clinician validation to ensure that they accurately reflect the original trial requirements. Once validated, the eligibility criteria are stored for deterministic evaluation.

Using the validated structured patient profiles and the validated logical representations of clinical trial criteria, the system performs a deterministic comparison to identify eligible patients. This process does not rely on probabilistic inference and ensures transparent and reproducible eligibility decisions.

The system outputs a list of eligible patients, as well as a detailed report explaining the reasons for patient exclusion and inclusion. These results are made available to clinicians through the system interface, supporting transparency, traceability, and informed decision-making.

All that was described can be seen on the Figure 1.2.

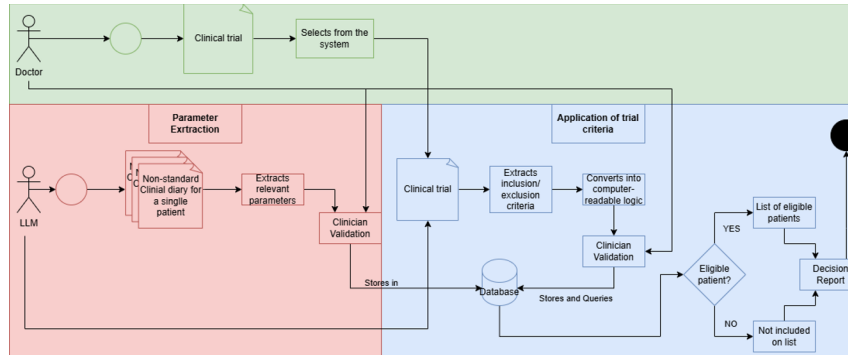


Figure 1.2: Proposed solution workflow

By introducing structured data, validation checkpoints, and deterministic evaluation, the proposed workflow significantly reduces manual workload, minimizes error propagation, and streamlines the clinical trial enrollment process, while preserving full clinician control over all critical decisions.

At a conceptual level, the proposed system supports clinicians by transforming unstructured clinical diaries into structured patient profiles and by enabling the deterministic comparison of these profiles with clinical trial eligibility criteria. Through this process, the system assists in identifying potentially eligible patients and in monitoring clinical outcomes, while preserving full clinician control. Importantly, the solution is designed strictly as a clinical decision-support system and does not perform autonomous clinical decision-making.

## 1.4 Contributions

The main contributions of this dissertation can be summarized as follows:

- The design and implementation of a clinical decision-support system capable of transforming unstructured oncology clinical diaries into structured patient profiles using Large Language Models.
- A semi-automated pipeline for extracting, normalizing, and validating clinical trial inclusion and exclusion criteria from free-text trial documentation, including their translation into machine-readable logical rules.
- A deterministic patient-trial matching mechanism that evaluates structured patient profiles against validated eligibility criteria, ensuring transparent, reproducible, and auditable eligibility decisions without reliance on probabilistic inference at the decision stage.
- The integration of clinician-in-the-loop validation at multiple stages of the pipeline, ensuring clinical correctness and interpretability.
- An applied evaluation of Large Language Models in a real hospital context, highlighting their potential and limitations when applied to unstructured clinical data in oncology departments.

## **1.5 Structure**

The remainder of this document is organized as follows.

Related Work presents an analysis of the state of the art relevant to the problem addressed in this dissertation.

Methodology and Planning details the methodological approach adopted for the development of the system, including the project management framework employed. It also describes the planning and management of the project, the tasks carried out during its development, and their temporal organization, illustrated through a Gantt diagram, as well as the activities planned for the next semester. Furthermore, it identifies the main risks associated with the project and outlines the mitigation strategies defined to ensure its successful completion.

Approach describes the proposed solution in detail and the design decisions that guided its implementation.

Requirements presents the requirements analysis, discussing the needs, constraints, and expectations of the main stakeholder, as well as the formal specification of the system's requirements.

Architecture introduces the C4 model for the software architecture, formalizing it from the context level down to the component level.

Experimental Setup defines and guides the experiments and exploratory analysis planned for the next semester.

Conclusion provides a brief summary of the work completed and the activities to be carried out in the following semester.

# Chapter 2

## Methodology and Project Planning

In this chapter, the methodology adopted for the development of the project is formally presented. The structure of the sprints, the management tools used to monitor progress, and the overall workflow are described in detail. Additionally, a clear overview of the work completed during the first phase is provided, supported by an illustrative diagram. This chapter also outlines the planning for the upcoming phase of the project and concludes with an analysis of the main project risks and the corresponding mitigation strategies.

### 2.1 Methodology

#### 2.1.1 Sprint structure and ceremonies

The workflow adopted during the first phase of the project followed a sequential structure of tasks, supported by weekly status update meetings. These meetings served as a space for presenting the work completed throughout the week and for receiving guidance regarding the upcoming tasks. Each meeting acted as a validation checkpoint, ensuring that the project remained aligned with the expected direction and objectives. The advisor was consistently present and actively involved in reviewing and questioning the work produced. Tutor Adelino and Dr. Margarida also participated, ensuring that the project remained aligned with the client's needs and providing additional guidance whenever necessary.

Although the project incorporated agile principles, particularly the possibility of revising and refining previously developed documents, the overall progression maintained a sequential flow. The work began with the definition of the problem and the study of related work and background knowledge, and only then moved toward the detailed specification LLM's workflows. Throughout the semester, additional meetings with the stakeholder were held whenever needed, either to validate progress or to clarify doubts that emerged during development.

### **2.1.2 Management Tools**

To support the management of the project, several tools were employed. Trello and Google Drive were used for task tracking and documentation, ensuring that each task had a defined start date and an expected completion date. This approach helped maintain organization and prevented work from falling behind schedule.

## **2.2 Work Conducted in the First Semester**

During the first phase of this thesis, a substantial amount of work was carried out, primarily focused on formalizing documentation and specifications to guide the development of the project in the following semester. The initial weeks of the project were mainly dedicated to defining the project scope and studying the relevant related work.

While the project definition was being refined, an initial study on parameter extraction from unstructured clinical diaries was already underway. Once the project scope was clearly defined, the topic of patient–clinical trial matching was explored in more depth, leading to the creation of the related work document. In parallel with this second phase of the related work, a study on Large Language Models was conducted, focusing on their evolution and capabilities. During this period, several meetings were held with the stakeholder to better understand the current clinical workflow and the underlying motivation for the project.

Based on these insights, the LLM-based workflow of the proposed solution was conceived and defined. Following this, three refined modules were established: the unstructured patient parameter extraction workflow, the clinical trial matching workflow, and a study on how patient care monitoring should be supported within the proposed solution. To provide a tangible sense of progress to the stakeholder, interface mockups were developed.

As a result of these activities, the overall software architecture was formally documented. Throughout this entire phase, the specification of requirements was continuously updated, as it evolved alongside the project definition and required constant refinement. Finally, a generalized structural analysis of both clinical diaries and clinical trial documents was performed. The final segment of this first phase was fully dedicated to the formalization of the thesis document.

All the activities described above are summarized in the Gantt diagram presented in Figure 2.1.

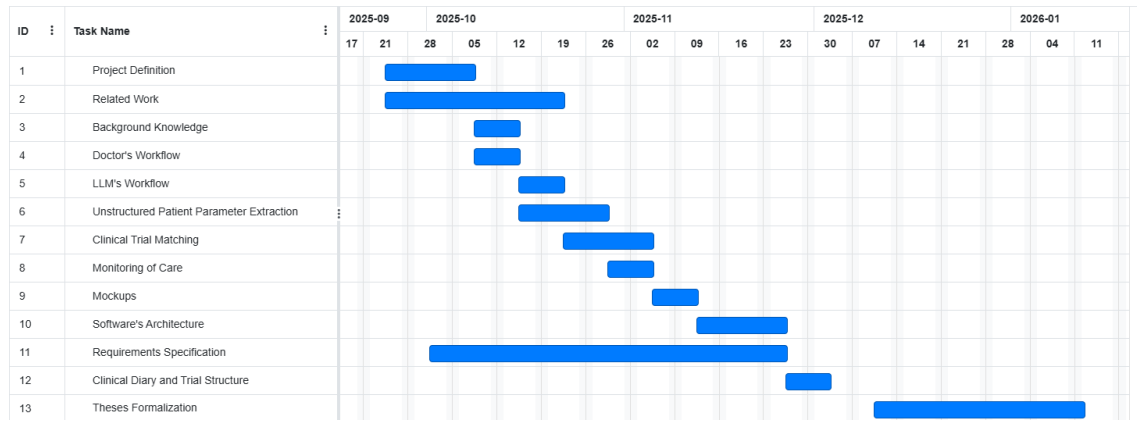


Figure 2.1: Gantt Diagram of first semester

## 2.3 Work Plan for the Second Semester

During the second semester, the implementation of the software system is planned. This phase is expected to begin with the finalization and validation of the current system requirements, allowing the software development process to start.

Following this, an extensive experimentation phase with Large Language Models and prompt strategies will be conducted. This experimentation will focus on unstructured patient information extraction, patient-to-trial matching, and the conversion of clinical trial criteria into logical rules, with the objective of identifying the most effective configurations to be integrated into the system.

In parallel with the experimentation phase, backend and user interface development will be initiated. As these two activities approach their final stages, system integration will begin, consisting of linking every component, UI, backend and LLM's, conducting full workflow tests with realistic data. With this the deployment preparation will begin, including the setup of a production-like environment and the execution of functional and integration tests on the application.

The final stage of this semester will be dedicated to the completion of the thesis document. Documentation will be produced continuously throughout this phase to ensure that all development and experimentation activities are properly recorded.

The planning for this semester is illustrated in the Gantt diagram shown in Figure 2.2.

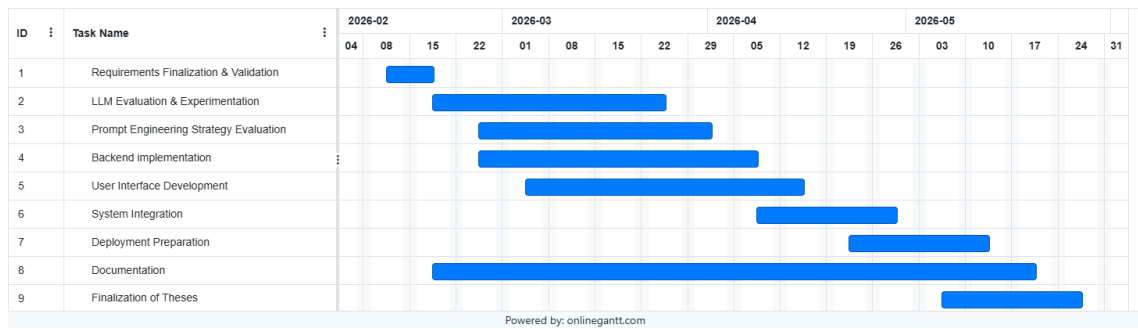


Figure 2.2: Gantt Diagram of second semester

## 2.4 Risk Management

The development of this project was subjected to a set of inherent risks to the exploratory nature and technological context associated. This risk are related with technical factors, limitations of resources or external dependencies, being able to impact the project's success and progress.

The following risk table has the below legend for its columns:

- ID, unique identifier for the Risk
- Risk, the risk itself, formalized on a affirmation → consequence manner
- I, impact that the risk will have on the project if it happens, this metric is evaluated on a 1-3 scale, where 1 corresponds to a negligible risk and 3 to a critical effect
- P, probability of the risk happening, this metric is evaluated on a 1-3 scale, where 1 corresponds to a unlikely event and 3 to a highly likely event
- Sev, severity of the risk calculated as  $\text{Impact} \times \text{Probabilty}$ , making severity a 1-9 scale

Bellow there is a comprehensional table of the scales for the 3 last described columns, Table 2.1:

| Metric      | Class | Criteria                                                  |
|-------------|-------|-----------------------------------------------------------|
| Impact      | 1     | Negligible, minimal disruption, no effect on deliverables |
|             | 2     | Moderate, noticeable impact requiring corrective action   |
|             | 3     | Critical, severe impact threatening project success       |
| Probability | 1     | Very unlikely, low chance of occurring                    |
|             | 2     | Moderate, could occur during the project                  |
|             | 3     | Very likely, high chance of occurring                     |
| Severity    | 1–2   | Low, limited effect on project execution                  |
|             | 3–4   | Moderate, requires management attention                   |
|             | 6–9   | High, potential to compromise project success             |

Table 2.1: Risk Assessment Metrics and Classification Criteria

Bellow the Table 2.2 presents the main risks identified, along with their impact, probability of occurrence, severity and mitigation plan defined to bridge the risk.

| ID | Risk                                                                                                                                                                                                                                                                                               | I | P | Sev | MP                                                                                                                                                                                                                                                                                                                |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1 | Limited access to real clinical data for experimentation. Because medical records are subject to strict ethical, legal, and privacy constraints, obtaining large and diverse datasets of unstructured clinical diaries and patient profiles may not be possible.                                   | 3 | 2 | 6   | Synthetic clinical data will be generated to simulate realistic patient profiles and clinical trial scenarios. This synthetic data will be designed to reflect real-world distributions, clinical constraints, and oncology-specific characteristics such as staging, treatment timelines, and survival outcomes. |
| R2 | Dependence on Large Language Models to extract structured information from clinical diaries and to convert clinical trial criteria into logical rules may produce incorrect, incomplete, or ambiguous outputs, especially when dealing with complex medical terminology or poorly structured text. | 3 | 3 | 9   | Clinician validation steps for both extracted patient information and translated eligibility criteria must be mandatory. These checkpoints ensure that no automated decision is made without expert review.                                                                                                       |
| R3 | Large Language Models may struggle with highly specialized oncology terminology, ambiguous phrasing, or institution-specific writing styles, affecting the accuracy of patient parameter extraction and the translation of eligibility criteria.                                                   | 2 | 2 | 4   | Conduct preliminary tests with representative clinical content, refine prompt templates iteratively, and incorporate domain-specific examples to improve extraction consistency.                                                                                                                                  |
| R4 | Running large models locally may exceed available computational resources, leading to slow inference times or making certain experiments infeasible.                                                                                                                                               | 2 | 2 | 4   | Use optimized execution environments, select model sizes appropriate to the available hardware, and offload heavier experiments to cloud-based environments when necessary.                                                                                                                                       |
| R5 | The project and experimentation require clinician involvement to validate extracted parameters, assess the correctness of eligibility criteria translations, and ensure alignment with real clinical workflows. Limited clinician availability may delay validation cycles.                        | 2 | 1 | 2   | Schedule validation sessions in advance, define objective validation criteria, and maintain short, focused review iterations to minimize clinician workload.                                                                                                                                                      |

Table 2.2: Project Risks and Mitigation Strategies

# Chapter 3

## Related Work

In this chapter, the relevant scientific literature underpinning the project is reviewed. Each selected paper is briefly summarized, highlighting its main contributions, limitations, and results, thereby establishing a clear foundation for identifying the gaps addressed by the proposed solution. The chapter is organized around the two core components of the system: the extraction of unstructured clinical information using large language models (LLMs) and the matching of patients to clinical trials through LLM-based methods. For each of these areas, a concise synthesis is provided, outlining the key insights and requirements that inform the design of the proposed solution.

Before starting the study it is important to have well defined these concepts, since they will be of major importance for the following description:

- **Open-Source Models**, an open-source model, is a completely transparent model, whose architecture, source code, training data and weights are made available for public use, therefore, emphasizing collaboration and redistribution, [xen];
- **Open Models**, an open model is a publicly accessible machine learning model that supports transparency, reuse, and collaborative improvement. although it doesn't guaranty full rights for modification and redistribution, [lyz, 2025];
- **Closed/Proprietary Models**, this is a model which is owned by a company, using private training data and resources. They are not publicly available and can only be acces via API's, [pro].

### 3.1 Extraction of Unstructured Clinical Information through LLMs

The extraction of meaningfull information from unstructured clinical data such as electronic health records (EHRs) is a fundamental challenge in medical informatics. Recent studies have explored the potential of large language models (LLMs)

to automate and enhance this process, focusing on diverse tasks including entity recognition, summarization, and classification of clinical text. The following works provide an overview of the methods, results and limitations associated with these approaches

### 3.1.1 Evaluating LLM Training and Prompting Strategies

The paper [Vithanage et al., 2024] investigated various training and prompting methods for improving LLM accuracy in clinical contexts. Techniques such as zero-shot and few-shot prompting were tested in combination with Parameter-Efficient Fine-Tuning (PEFT) and Retrieval-Augmented Generation (RAG). The results showed that few-shot prompting outperformed zero-shot prompting when no additional techniques were applied, and that PEFT produced the largest performance gains, improving accuracy, precision, recall, and F1 by over 30% in the zero-shot setting. RAG offered smaller but still meaningful improvements, particularly in the few-shot configuration. Overall, PEFT proved to be the most impactful technique, offering strong guidance for future fine-tuning approaches to unstructured clinical text analysis.

### 3.1.2 Comparative Evaluation of LLMs for Information Extraction

The study [Vithanage et al., 2025] proposed a holistic evaluation framework for information extraction (IE) from free-text clinical notes. It systematically assessed different LLMs on Named Entity Recognition (NER) and summarization tasks using real-world unstructured data. In its optimal configuration, LLaMA 3.1-8B achieved high NER performance ( $\approx 88\%$  accuracy and 90% F1) and strong summarization quality ( $\approx 88\%$  F1) when combining RAG with few-shot prompting. The combined RAG setup (LangChain + LlamaIndex) consistently outperformed isolated pipelines, improving both NER and summarization metrics. Health domain specialized models showed only marginal advantages over general-purpose ones, with differences not reaching statistical significance. Despite strong scores, the study reported notable robustness challenges, with error rates rising from 38% to 49% under noisy conditions and false negatives representing the majority of errors. These findings highlight the need for domain adaptation and hybrid architectures to ensure reliable performance in sensitive medical settings.

### 3.1.3 Efficiency of LLMs Compared to Manual Review

In [Bhagat and Mackey], the efficiency of ChatGPT 3.5 was compared against manual data review for extracting clinical information from unstructured records. Despite being an out-of-the-box model, ChatGPT exhibited high specificity and sensitivity across several conditions—for example, it reached 1.00 sensitivity and 0.95 specificity in depression, 1.00 sensitivity and 0.94 specificity in heavy-smoker detection, and 0.85 sensitivity with 1.00 specificity for cancer-related information,

while performance dropped for family history of cardiac disease (0.58 sensitivity, 0.97 specificity). These results show that the model could reliably identify most clinically relevant concepts, although nuanced semantic distinctions remained challenging. The study also reported stable outputs across repeated runs and noted that traditional systems such as MedTagger previously achieved 0.98 sensitivity and 0.96 specificity for cardiac-risk extraction. Overall, the findings suggest that LLMs can outperform classical NLP pipelines that require extensive manual engineering, and that targeted fine-tuning and structured extraction rules could further enhance their reliability in clinical settings.

### **3.1.4 Large-Scale Model Benchmarking for Data Extraction**

The work [Ntinopoulos et al., 2025] assessed the performance of eighteen different LLMs using synthetic medical notes validated by specialists. The evaluation covered four entity-extraction tasks and five binary-classification tasks under a zero-shot setting. Output consistency was measured across repeated prompt iterations. Advanced LLMs such as Claude 3.0 Opus, Claude 3.0 Sonnet, Claude 2.0, and GPT-4 achieved overall accuracies above 0.98, with Claude 3.0 Opus reaching 0.995, and nine models achieving perfect accuracy (1.0) in entity extraction. In binary classification, Claude 3.0 Opus failed only 2 out of 250 tasks, while GPT-4 and Claude 2.0 failed 5 out of 250. These top models also showed strong detailed metrics—for example, Claude 3.0 Opus achieved an F1 of 0.992, and GPT-4 an F1 of 0.979—and high consistency, with Claude 2.0 reaching a Krippendorff’s alpha of 1.0 across all iterations. Overall, these results demonstrate that modern LLMs substantially outperform baseline systems such as RoBERTa (accuracy 0.742) and can reliably extract structured information from EHRs, reducing clinicians’ data-processing workload.

### **3.1.5 Extraction of Social Determinants of Health (SDoH)**

The study [Gu et al., 2025] explored whether open-source LLMs could effectively extract Social Determinants of Health (SDoH) from clinical text. Two pipelines, Default and Refined, were implemented, with the latter incorporating prompt engineering to mitigate hallucinations. Manual annotations by human reviewers served as gold standards, with inter-annotator agreement reaching 93%. Results showed that LLMs outperformed the traditional pattern-matching baseline, which achieved 77% overall accuracy but only 39% accuracy on explicit mentions. In contrast, openchat 3.5 was the only model to surpass the baseline across all nine SDoH categories, improving direct-mention accuracy by over 40% and outperforming the baseline Macro-F1 score of 0.53–0.54. Models such as Llama-2 performed significantly worse than the baseline, while the refined pipeline reduced hallucinations and invalid outputs. Quantized variants remained robust under a 2048-token limit, although none of the models detected SDoH elements missed by human reviewers. These findings highlight the importance of prompt design for improving accuracy in unstructured clinical text extraction without fine-tuning.

### 3.1.6 Oncology Information Extraction with Expert-Annotated Data

The [Sushil et al., 2024] paper introduced a new annotated dataset (CORAL) for evaluating LLMs in oncological information extraction. The corpus contained 40 progress notes annotated hierarchically by oncology specialists, capturing entities, attributes, and relationships. Three models: GPT-4, GPT-3.5-turbo, and FLAN-UL2, were tested in a zero-shot setting on extracting clinical narrative elements such as symptoms, test results, and treatment plans. GPT-4 achieved the best overall progression, with manual evaluation confirming 68% accuracy. Nevertheless, hallucinations and inconsistent causal reasoning persisted, indicating that even state-of-the-art models are not yet suitable for direct clinical application. Moreover, the study did not explore the potential benefits of prompt engineering, leaving room for future enhancement.

### 3.1.7 Summary

Across these studies, LLMs consistently demonstrate strong capabilities in extracting clinical information from unstructured data, frequently outperforming traditional NLP and rule-based systems. Techniques such as PEFT, RAG, and carefully engineered prompts emerge as key factors for improving accuracy and reliability. However, limitations persist in terms of hallucination control, contextual reasoning, and semantic nuance understanding. The need for domain-specific fine-tuning, evaluation frameworks, and interpretability remains central to ensuring trustworthy clinical deployment.

## 3.2 Matching Patients for Clinical Trials with LLMs

The automation of patient-trial matching represents a major frontier in clinical informatics, aiming to accelerate recruitment, reduce operational costs, and ensure equitable access to medical research. Traditional matching pipelines are labor-intensive and constrained by the unstructured nature of electronic health records (EHRs). Recent studies have demonstrated that large language models (LLMs) can address these challenges by interpreting clinical narratives and eligibility criteria, offering explainability and scalability through zero-shot and few-shot learning. This section presents a synthesis of recent works exploring LLM-based frameworks for clinical trial matching.

### 3.2.1 Zero-Shot Matching and the TrialGPT Framework

The paper [Jin et al., 2024] introduces TrialGPT, a modular framework employing LLMs for zero-shot patient-trial matching. It consists of three interconnected modules:

1. TrialGPT-Retrieval, which filters a large pool of trials by extracting key-

words from EHRs;

2. TrialGPT-Matching, which evaluates patient eligibility criterion-by-criterion, generating natural language explanations, localizing relevant EHR phrases, and producing an eligibility classification; and
3. TrialGPT-Ranking, which aggregates eligibility predictions at the trial level to rank potential matches.

Two ranking strategies were proposed: Linear Aggregation, based on direct percentage computations for inclusion and exclusion criteria, and LLM Aggregation, which leverages LLMs to infer global relevance and eligibility. TrialGPT-Retrieval recalled 90% of relevant trials using only 6% of the total dataset, with the generated keywords surpassing those manually produced by experts. TrialGPT-Matching achieved an 87.3% accuracy, nearly matching human specialists (0.887-0.900), and demonstrated an F1 score of 88.6% in identifying relevant phrases. The outperforming competing systems by 43.8% in trial inclusion and exclusion classification. The study also estimated a 42.6% reduction in screening time, establishing TrialGPT as an efficient foundation for clinical trial automation.

### 3.2.2 Retrieval Pipelines and LLM-Based Re-Ranking

In [Rybinski et al., 2024], a retrieval-based pipeline was developed to match patients with suitable clinical trials. The process began with query formulation, representing each patient’s profile as a query, using either a template-based or free-text representation generated by GPT-3.5-turbo. A BM25 model performed initial retrieval of 1,000 candidate trials, indexing both individual and aggregated fields of trial descriptions. This was followed by a re-ranking stage, where several strategies were applied:

- SCT-BERT, a fine tuned BERT variant specialized for clinical trial text;
- TCRR, inspired by curriculum learning, which first learns topical relevance and then fine-tunes on true eligibility;
- LLM-based embeddings, using semantic similarity between the patient and trials texts;
- LLM-based relevance scoring, where GPT-3.5-turbo directly produces numerical ranking scores;
- LLM-based filtering, applying binary yes/no filtering to the top candidates.

The results showed that fine-tuned LLM-based re-ranking achieved the highest retrieval precision, with the BM25-ES + TCRR + fine-tuned GPT pipeline reaching 0.777 nDCG@10, 0.697 P@10, and 0.783 RR, clearly outperforming the BM25 baseline (0.619 nDCG@10, 0.330 P@10). Zero-shot filtering with GPT achieved 0.720 nDCG@10 and 0.592 P@10, while GPT-4o delivered competitive zero-shot results, including 0.785 nDCG@10 and 0.844 RR, though with substantially higher

latency. Latency varied widely across components, from 0.15 s for BM25 to 46.48 s for LLM re-ranking, with GPT-4o CoT reaching 242 s. The study highlighted the effectiveness–efficiency trade-off, noting that fine-tuning required around 10,000 training samples and increased latency roughly 15× for a 12% gain in nDCG@10. Overall, PLM-based re-rankers offered a more cost-efficient balance, while LLMs remained computationally demanding despite their superior matching quality.

### 3.2.3 Ontology-Enhanced Prompt-Baed Matching

The work [Rahmanian et al.] proposed an approach combining LLMs with structured ontologies such as SNOMED CT to refine eligibility extraction. Using GPT-3.5-turbo in a deterministic configuration (temperature = 0), the system automatically annotated clinical records with MedCAT, linking text to SNOMED CT and UMLS terminologies. Phrases containing relevant concepts were extracted to reduce input length while preserving essential information. Each eligibility criterion was converted into a question-based prompt for binary classification ("yes" or "no"), forming a zero-shot evaluation pipeline.

The model achieved micro and macro F-measures of 0.9061 and 0.8060, respectively placing it among the top-performing systems on the n2c2 dataset. without summarization, the performance dropped considerably (micro: 0.7126; macro: 0.5277), underscoring the value of preprocessing and semantic abstraction. Limitations included high false-negative rates caused by non-standard abbreviations, ontology gaps (e.g., missing synonyms and street drug names), and class imbalance in certain criteria. Despite these challenges, the study demonstrated the viability of ontology-enhanced prompting for improving interpretability and precision in patient eligibility classification.

### 3.2.4 Distilling Open-Source Models for Patient-Trial Matching

The paper [Nievas et al., 2023] presented the first systematic evaluation of open-source versus proprietary LLMs in this domain. The study compared GPT-3.5, GPT-4, and LLAMA-2 (7B, 13B, 70B) models, with a focus on refining open-source systems to rival proprietary performance. Each model performed multi-step reasoning: generating chain-of-thought explanations, identifying evidence phrases, and categorizing patients as "included", "excluded" or "no relevant information". The trial ranking metric was computed as the difference between inclusion and exclusion fulfillment percentages.

To improved open-source adaptability, a synthetic dataset of 2,000 patient-trial pairs was generated by GPT-4, including explanations and evidence. LLAMA variants were fine-tuned using supervised loss on this dataset, resulting in the Trial-LLAMA 70B model. Evaluation metrics included NDCG@10, AUROC, and Criterion-Level Accuracy (CLA) for explicit and implicit reasoning, along with Evidence Faithfulness (precision, recall, and F1). GPT-4 remained the top-performing out-of-the-box model, while Trial-LLAMA 70B achieved performance parity with GPT-3.5 and even exceeded it in criterion-level accuracy after fine-tuning. The

results confirmed that open-source models can achieve near-proprietary performance with task-aligned fine-tuning.

However, implicit reasoning remained challenging across all models, and the reliance on GPT-4-generated training data introduced potential bias. Furthermore, improvements in criterion-level accuracy did not always translate to higher trial-level ranking performance, revealing a non-linear relation between component accuracy and end-task success.

### 3.2.5 End-to-End Matching with Real-World Validation

The study [Gupta et al., 2024] represents a significant advance, as it offers the first empirical end-to-end validation of a trial matching system using real-world unstructured EHRs. The PRISM framework formulates eligibility matching as a question-answering problem, where inclusion and exclusion criteria are decomposed into simplified question with possible answers ("yes", "no", or "N/A"). Logical structures such as Disjunctive Normal Form (DNF) are employed to manage multi-question criteria.

PRISM introduces OncoLLM, a fine-tuned LLM based on Qwen-1.5 (14B parameters), specialized in oncological data and augmented with Chain-of-Thought reasoning. To handle long EHRs, the pipeline includes a chunking and retrieval module using spaCy for tokenization and chronological arrangement of patient notes. The final decision-making follows a logical tree, and probabilistic evaluation is used for missing data. Trials are ranked using a Weighted Tier method prioritizing clinically significant criteria.

OncoLLM achieved 63% criterion-level accuracy, approaching GPT-4's 68%, while outperforming GPT-3.5-turbo and similarly sized models. It also excelled in both patient-centric and trial-centric rankings, with 95% of top-3 recommendations judged clinically appropriate and exclusive use of unstructured data (omitting structured records such as lab results), dependence on imperfect retrievers, and annotation inconsistencies among human experts. These findings underline the need for hybrid systems combining structured and unstructured data with consistent expert validation.

### 3.2.6 Scalable Zero-Shot Matching Systems

Finally, [Wornow et al., 2024] demonstrated the feasibility of scaling patient-trial matching through a zero-shot architecture. The system evaluated patient eligibility by comparing unstructured clinical histories to trial criteria using models such as GPT-4, LLAMA-2-70B-32k, and Mixtral-8x7B. Lightweight embedding models (BGE and MiniLM) were used to preliminary retrieval in a two-stage pipeline. Four prompting strategies were explored: All Criteria, All Notes (ACAN); All Criteria, Individual Notes (ACIN); Individual Criteria, All Notes (ICAN); and Individual Criteria, Individual Notes (ICIN); With ACIN identified as the most cost-efficient.

Using GPT-4 under the ACIN strategy, the system achieved state-of-the-art results on the n2c2 benchmark, with a Micro-F1 of 0.93 and Macro-F1 of 0.81, surpassing prior rule-based systems and other LLMs. It also achieved dramatic cost reductions, screening 86 patients in two hours at only \$1.55 per patient. Clinician evaluation confirmed 97% coherence in model explanations. However, limitation included dataset simplification (English-only, limited criteria coverage), prompt dependence, and potential hallucinations. Privacy and compliance concerns also persist for proprietary models under healthcare regulations such as HIPAA.

### 3.2.7 Summary

Across all studies, LLMs demonstrate exceptional potential in automating patient-to-trial matching by leveraging zero-shot reasoning, retrieval-augmented architectures, and prompt-based interpretability. Fine-tuning and ontology integration notably improve performance, while hybrid retrieval pipelines balance speed and accuracy. Open-source models such as Trial-LLAMA and OncoLLM show that domain-specific fine-tuning can achieve or surpass proprietary benchmarks.

However, key challenges persist: high computation costs, sensitivity to prompt design, limited robustness in implicit reasoning, and difficulties in establishing reliable ground truths. Furthermore, real-world deployment must address privacy, interpretability, and integration with structured clinical data. Future research should focus on hybrid multimodal systems, explainability, and the harmonization of performance with efficiency to achieve clinically trustworthy deployment.

## 3.3 Conclusion

Having considered the evidence across the reviewed studies, our approach aims to employ LLM-based parameter extraction from unstructured clinical diaries while explicitly addressing the well-documented risk of hallucinations. To ensure reliability, every model output entering the application will undergo clinician validation. In addition, we adopt a trial-to-patient matching strategy in which clinical trial criteria, although expressed in natural language, are treated as rule-based constraints. These criteria must therefore be translated into machine interpretable logical rules and subsequently validated by clinical experts before being used in a deterministic matching pipeline. Overall, our methodology leverages the strengths of LLMs while mitigating their limitations through a clinician-in-the-loop design.

# Chapter 4

## Approach

In this chapter, a clear formalization of how the solution was designed is presented, taking into account its two main components: the extraction of unstructured clinical information using large language models (LLMs) and the matching of patients to clinical trials through LLM-based methods. Furthermore, a generalization of the two key document types involved in the system, clinical trials and patient diaries, is provided. Finally, this chapter includes a justification of the tools considered for use in the next phase of the project.

### 4.1 Method Overview

As described in previous sections, this project aims to reduce the workload of clinicians by leveraging Large Language Models (LLMs) to support the structuring of patient profiles from free-text clinical diaries and the identification of potentially eligible patients for clinical trials. Given the non-deterministic nature of LLMs and the critical clinical context, all automated steps require expert validation by the clinician.

To support these interactions, the system incorporates a user interface that allows clinicians to submit data, review intermediate results, validate outputs, and inspect final decisions. This interface acts as the main interaction point between the clinician and the underlying LLM-based processing pipeline.

At a methodological level, the system pipeline is divided into two main phases:

- Phase 1: Extraction of Hyper-Relevant Parameters
- Phase 2: Application of Clinical Trial Criteria

## 4.2 Hyper-Relevant Parameter Extraction

The goal of the first phase is to transform free-text, unstructured clinical information into normalized and structured data. The input to this phase consists of clinical diaries written in free-text format, which are selected by the clinician through the system interface. The process follows these steps:

1. Text Pre-processing: Clinical diaries are retrieved from SClínico and their textual content is extracted and prepared for processing.
2. Information Extraction: The LLM receives the extracted text along with a carefully crafted prompt and is instructed to identify and extract hyper-relevant clinical parameters.
3. Validation and Structured Output Generation: The LLM-generated output is presented to the clinician through the interface for expert validation and possible correction. Once approved, the validated information is stored in the database in a structured format (e.g., JSON) for subsequent use.

Upon completion of this phase, a structured database containing the validated key clinical parameters for each patient is obtained.

## 4.3 Application of Clinical Trial Criteria

The goal of the second phase is to evaluate the structured patient profiles against the inclusion and exclusion criteria of a given clinical trial. The inputs to this phase are the structured patient profiles produced in the previous phase and the clinical trial documentation, both of which are submitted and managed through the system interface.

The process follows these steps:

1. Text Pre-processing: Clinical trials must undergo a pre-processing, for the removal of any irrelevant sections to the proposed solution;
2. Criteria Extraction and Translation: Inclusion and exclusion criteria are extracted from the clinical trial documentation and translated into computational logic (e.g., Age > 18) using an LLM;
3. Criteria Validation: The extracted logical representations are presented to the clinician for expert validation to ensure that they accurately reflect the original trial requirements;
4. Deterministic Eligibility Evaluation: Once both patient profiles and trial criteria are structured and validated, a deterministic assessment is performed, producing a categorical eligibility result: Eligible, Not Eligible, or Inconclusive;

Following this process, the system generates two main artifacts, which are made available to the clinician through the interface:

- **Eligible Patient List:** A list of patients who satisfy all the clinical trial criteria and are therefore considered eligible.
- **Exclusion Report:** A detailed report specifying, for each non-eligible patient, which criteria were not satisfied, supporting transparency and clinical justification.

## 4.4 Clinical Documents generalization

Two main types of clinical documents will be manipulated in this solution, clinical diaries and clinical trials. Below it is possible to find a generalized template of how both clinical documents are organized in practice, independent of therapeutic are.

The generalized template refers to a canonical data model developed by identifying recurring information patterns across both clinical diaries and trials. It acts as a standardized schema that provides a predefined slots to normalize this information into a logic-based format. By defining these parameters upfront, the template serves as a structural guide for the Large Language Model (LLM) via prompt engineering, ensuring consistent data extraction and the deterministic conversion of clinical criteria into machine-readable rules.

These abstract representations were made and structured having as base real documents provided by the stakeholder to use as example.

Starting with the Clinical Trial protocol. Although clinical trial protocol are written as narrative documents, they follow a well-defined structural organization as can be seen the example bellow. Among these sections, eligibility criteria represent the only on that directly decides the patient selection. While expressed in natural language, inclusion and exclusion criteria are inherently constraint-based, specifying admissible values, thresholds, or conditions over patient attributes.

This structure allows for a systematic separation between descriptive trial information and eligibility logic. By isolating this eligibility within the trial template, the proposed methodology can safely apply Large Language Models to extract and normalize patient selection criteria without affecting other trial metadata. The extracted criteria can then be deterministically translated into machine-readable rules, allowing automated comparison against structured patient records.

The conversion process goes as follows:

- “Aged  $\geq 18$  years” `age  $\geq 18$`
- “ECOG PS 0 or 1” `ECOG_PS in 0,1`
- “HLA-A2 positive” `HLA_A2 == true`

- “Creatinine clearance > 30 ml/min” CrCl > 30
- “No brain metastasis” brain\_metastasis == false
- “Progressed after 24 weeks of ICI” weeks\_on\_ICI >= 24 AND progression == true

This template can be found below on the Figure 4.1

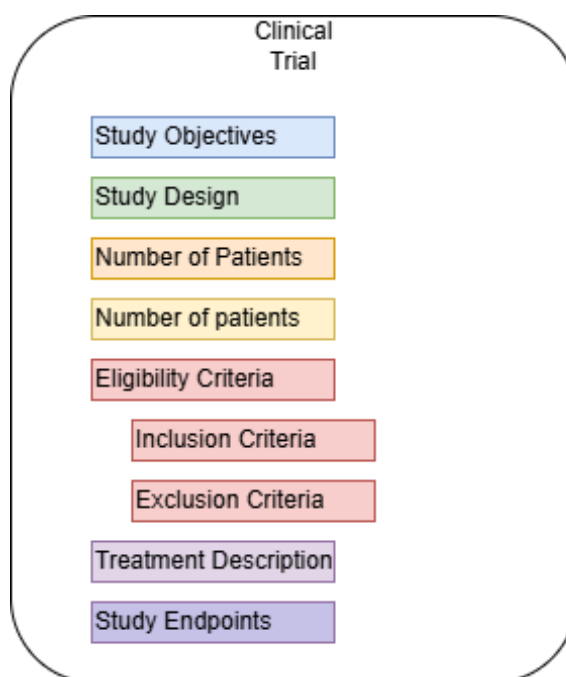


Figure 4.1: Clinical Trial Abstraction

The design of the extraction pipeline is fundamentally driven by a structured template derived from four representative clinical diaries provided by stakeholders. By establishing this foundational template, the system ensures that the diverse nature of free-text clinical notes is mapped to a consistent format. The selection of specific extraction parameters—including age, ECOG-PS, diagnosis details, molecular status, staging, and longitudinal treatment dates—serves as a targeted guidance system. This prevents "information overload" for the model and focuses the computational resources of the Large Language Model (LLM) on high-value clinical data points required for trial matching and survival analysis.

This template can be found below on the Figure 4.2

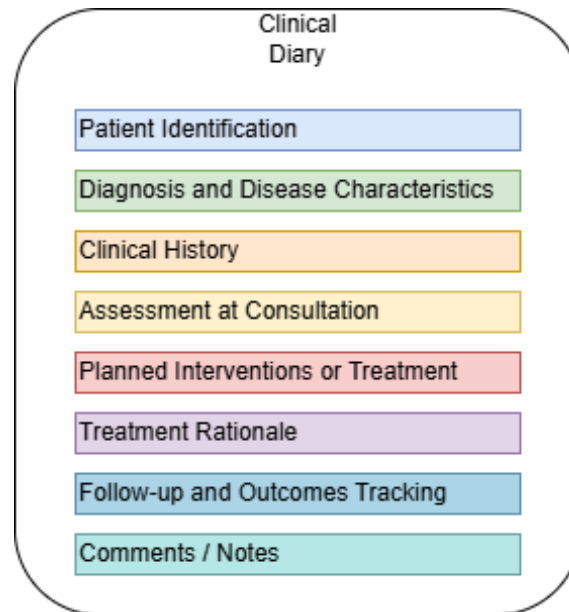


Figure 4.2: Clinical Diary Abstraction

## 4.5 Choices and Tools

The selection of methodologies and tools employed in the current project directly arise from the nature of the available clinical data, defined functional requirements and the necessity of ensuring reproducibility, auditability and clinical rigor. The proposed solution combines Natural Language Processing (NLP) techniques, classical static methods and a robust data structure, allowing the transformation of unstructured clinical information in actionable knowledge for decision support.

### 4.5.1 Usage of Language Models

The choice of LLMs as the core of the extraction and normalization process is justifiable because of the high linguistic variability present in the clinical diaries. These documents present:

- Free writing and not standardized;
- Orthographic errors and abbreviations;
- Lack of formal structure

Traditional NLP tools based on rules or dictionaries can not efficiently deal with this heterogeneity. LLMs on the other hand can:

- Interpret clinical text;

- extract entities and relations without previous annotations;
- significantly reduce manual effort from the clinicians

That's why LLMs were chosen as the most adequate methodology for the compliance of requirements FR3.2 - Extract structured info-FR3.3 - Normalize text and FR1.1 - Extract clinical trial criteria-FR1.2 - Convert Criteria into logic.

## 4.5.2 JSON structure and PostgreSQL

The JSON format was chosen as the representation of the LLM's output because it leverages really well structure and readability (FR1.2 - Convert Criteria into logic and FR3.2 - Extract structured info). Since every extracted profile must later be reviewed and validated by a clinician (FR1.3 - Validation of inclusion and exclusion criteria and FR3.4 - Validation of extracted parameters), it is essential that the output remains human-readable and easy to interpret. JSON naturally supports this, with its key-value structure making the information explicit, organized, and simple to modify whenever the clinician needs to correct or complete a field. At the same time, JSON is simple to manipulate programmatically, which is important for the downstream steps of normalization, validation, and storage. Its structured nature also simplifies the process of persisting these objects in the database, ensuring that each field can be mapped cleanly into the corresponding schema without ambiguity.

Regarding the database, PostgreSQL was selected primarily due to familiarity with the tool, but this choice is also supported by its technical strengths. PostgreSQL is a mature, well tested relational database system known for its robustness and reliability qualities that are essential when dealing with clinical data. It offers strong guarantees in terms of data integrity, transactional safety, and recovery mechanisms in case of corruption or unexpected failures.

## 4.5.3 Conversion of Clinical Criteria and Deterministic evaluation of patients

The conversion of a clinical trial's inclusion and exclusion criteria into logical rules (FR1.2 - Convert Criteria into logic) requires two complementary components: a semantic interpretation of the free-text criteria, performed by an LLM; and an objective and deterministic evaluation of each patient against those rules. This separation is fundamental, as it prevents the introduction of probabilistic behaviour into a clinical decision making process and maximizes reproducibility. By ensuring that the LLM is responsible only for interpreting the text and not for making the final decision, the system maintains full control over how each rule is applied.

The eligibility decision (FR1.4 - Compare criteria with database) is therefore executed through deterministic logic rather than through a Machine Learning model.

Clinical criteria are explicit rules, and as such the decision must be entirely explorable and auditable. Clinical environments demand complete traceability, meaning that every step of the decision process must be transparent and reproducible. This deterministic evaluation, combined with clinician validation, guarantees that all inclusion criteria are satisfied, no exclusion criterion is violated, and that the final result is strictly binary and consistent across repeated evaluations.

#### 4.5.4 Pipeline

The system architecture follows a modular approach for both the patient parameter extraction and the conversion of clinical trial criteria, with the latter not requiring the Normalization step. The overall pipeline can be summarized as:

Extraction → Normalization → Validation → Storing → Evaluation

This methodology was chosen because it facilitates the evolution of the system, allowing components (e.g., the LLM) to be replaced without the need to rewrite the entire pipeline. It also ensures full traceability through logging, and guarantees that only validated data is used in downstream processes such as the quality-of-care analysis. This modularity is essential for clinical systems, which require robustness, auditability.

# Chapter 5

## Requirement Analysis and Theme

In this chapter, a clear formalization of the stakeholders' needs and constraints is presented, establishing the foundation for the system's functional and non-functional requirements. Additionally, the three core workflows of the solution are described in detail, automated structured data extraction, identification of eligible patients for clinical trials, and monitoring of care, each supported by an illustrative diagram. Together, these elements define the thematic scope of the project and outline the essential processes that the proposed system must support.

### 5.1 Context and Stakeholder Needs

As previously discussed, the current oncology setup is far from standardized. Clinical diaries are written in free-text and vary widely between clinicians, increasing the cognitive load on oncologists. Patient-trial matching is still performed manually, and relevant information is dispersed across multiple unstructured notes, making the process inefficient and error-prone.

This situation increases the likelihood of screening errors, which may lead to financial losses if the hospital fails to meet recruitment targets. These limitations highlight a major workflow problem and motivate the need for a software pipeline capable of connecting unstructured patient data with the manual screening process, reducing errors and preventing unnecessary costs.

The primary stakeholders of this project are the oncologists at ULS Coimbra, particularly Dra. Judy. Through several meetings, a set of key needs were identified:

- Structured patient profiles, since the reliance on free-text documentation creates variability and makes clinical-trial screening slow and error-prone. A tool capable of extracting relevant parameters and organizing them into structured profiles is essential to simplify this workflow.
- Explainable eligibility decisions, as clinicians must verify that each logical rule reflects the trial requirements and understand why a patient was in-

cluded or excluded. Even with deterministic matching after LLM extraction and clinician validation, transparency and auditability remain mandatory.

- Clinician validation, which is critical to ensure correctness of all LLM-generated outputs. Given the stochastic nature of LLMs and their susceptibility to hallucinations, human oversight is required to prevent erroneous information from propagating through the system.
- Outcome monitoring, enabling clinicians to evaluate treatment effectiveness over time and assess both clinical outcomes and quality of care.

Beyond these functional needs, the system must comply with strict constraints inherent to clinical environments. These constraints shape the requirements analysis and must be considered throughout system design.

First, data privacy is a fundamental restriction. Clinical records contain highly sensitive information and must be handled according to ethical and institutional policies. The system must therefore ensure controlled access, secure storage, and full traceability of all operations.

Second, auditability is essential. Systems used in clinical contexts must produce verifiable and justified results. All processes influencing trial eligibility must be transparent and reproducible, requiring retention of intermediate results, decision outcomes, and clinician interventions for retrospective analysis.

Lastly, the non-deterministic nature of LLMs introduces an additional constraint. Although effective for processing unstructured text, LLMs may generate inconsistent or incorrect outputs. To mitigate this, the system incorporates human in the loop validation, ensuring that all extracted information is clinically reviewed before use. LLMs are therefore employed strictly as decision-support tools, assisting in extraction and interpretation while all clinically relevant decisions remain under clinician control. This preserves authority, ensures accountability, and minimizes unintended clinical impact.

Taking into account the identified needs and constraints, three main categories of system requirements were defined

## 5.2 Automated Structured Data Extraction

This first category focuses on converting free-text clinical diaries into structured patient records. The workflow begins with the manual selection of one or more clinical diaries, following a pre-processing step of the document, for more information refer to Appendix FR0.2 - Pre-process clinical documents, which are submitted to an extraction pipeline based on Large Language Models. This pipeline applies deterministic prompting strategies to map essential clinical parameters such as: age or date of birth, ECOG-PS, primary diagnosis, date of diagnosis, molecular status, cancer stage (TNM or equivalent), treatment history, and corresponding temporal information into a structured format following a strict JSON

schema. To ensure data quality, a semantic normalization phase is applied to correct terminological inconsistencies without altering the underlying clinical meaning. The extracted data is then subject to clinician validation, allowing review and correction prior to persistence. Finally, the validated structured profiles and corresponding change logs are stored in a PostgreSQL database, ensuring traceability and integrity for downstream analyses.

For better visualization below on Figure 5.1 is the formalized diagram for this module.

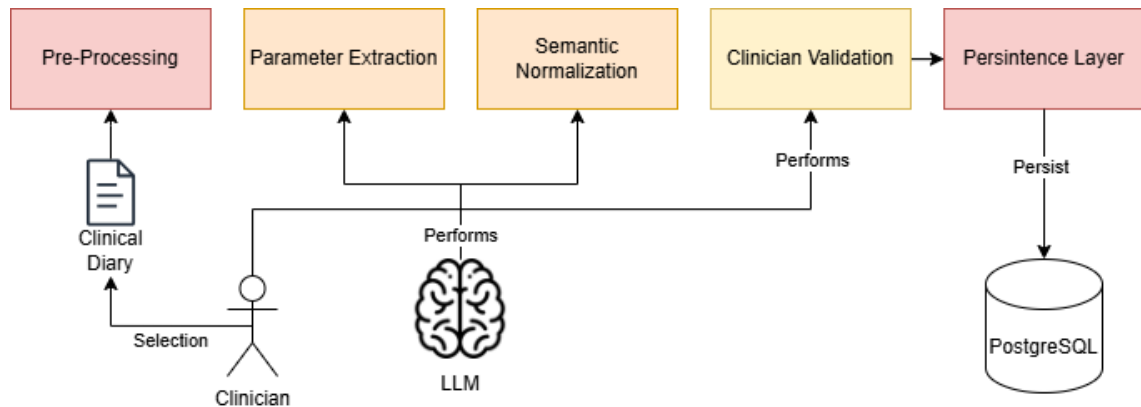


Figure 5.1: Parameter Extraction Workflow

Everything described above can be seen as requirements on the below table 5.1, for further specification refer to appendixes Automated Structured Data Extraction and Oyster Level Requirements.

| ID    | Requirement                        | Description                                                                                                                                                                                                                                                                                                          |
|-------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR3.1 | Select diaries for extraction      | The system must allow one or multiple diaries to be extracted.                                                                                                                                                                                                                                                       |
| FR3.2 | Extract structured info            | The LLM must extract patient data from free-text diaries, including: Age / Date of birth; ECOG-PS; Diagnosis; Date of diagnosis; Molecular status; Stage; Date of treatment; Treatments (First, Second); Start date; End date; Control. If a field does not exist in the clinical diary, it must be defined as None. |
| FR3.3 | Normalize text                     | The system must correct typographical and linguistic inconsistencies.                                                                                                                                                                                                                                                |
| FR3.4 | Validation of extracted parameters | After the LLM extracts the parameters, the doctor can edit and validate these data.                                                                                                                                                                                                                                  |
| FR3.5 | Store structured profiles          | Each structured patient record must be stored in the local database.                                                                                                                                                                                                                                                 |

Table 5.1: Automated Structured Data Extraction - Functional Requirements

### 5.3 Identification of Eligible Patients for Clinical Trials

This second category, aims to automate the matching between structured patient profiles and clinical trial criteria. The process begins with the submission of clinical trial documentation, following a pre-processing step of the document, for more information refer to Appendix FR0.2 - Pre-process clinical documents, from which inclusion and exclusion criteria are extracted using an LLM and subsequently converted into structured logical rules. To ensure clinical accuracy, these rules must undergo mandatory clinician validation before being stored. Once validated, the system performs a deterministic evaluation, comparing structured patient data against the trial's criteria. Unlike the extraction phase, this matching process does not rely on probabilistic inference, resulting in a strict binary eligibility decision. The system then generates an eligibility report, including a natural language justification for each decision, facilitating clinical interpretation. All results, including exclusion reasons, are persisted for audit purposes.

For better visualization below on Figure 5.2 is the formalized diagram for this module.

Everything described above can be seen as requirements on the below table 5.2, for further specification refer to appendixes Identification of Eligible Patients for

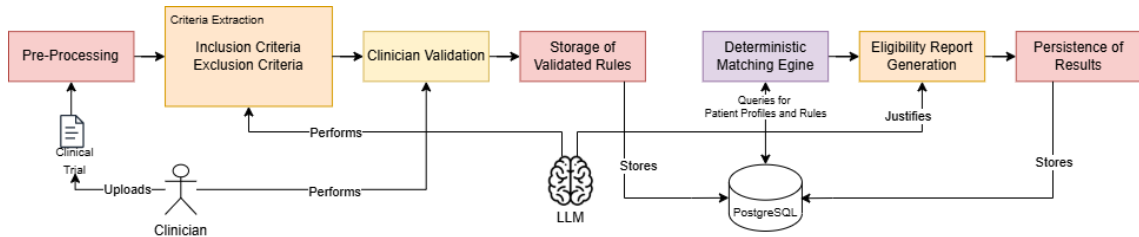


Figure 5.2: Patient-Trial Matching workflow

Clinical Trials and Oyster Level Requirements.

| ID    | Requirement                                    | Description                                                                                                                                                              |
|-------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR1.1 | Extract clinical trial criteria                | The LLM must extract inclusion and exclusion criteria from a clinical trial.                                                                                             |
| FR1.2 | Convert criteria into logic                    | The extracted text must be converted into logical expressions (e.g., Age > 18).                                                                                          |
| FR1.3 | Validation of inclusion and exclusion criteria | After the LLM extracts the inclusion and exclusion criteria, a second phase must occur in which the doctor validates these results in order for patients to be selected. |
| FR1.4 | Compare criteria with database                 | The system must automatically evaluate patient data against the logical criteria of clinical trials.                                                                     |
| FR1.5 | Generate eligibility report                    | The system must output a list of eligible patients and provide justification for excluded patients.                                                                      |
| FR1.6 | Saving included patients                       | After a patient is considered eligible for a trial, they should be paired with it and this information must be persisted.                                                |

Table 5.2: Identification of Eligible Patients for Clinical Trials – Functional Requirements

## 5.4 Monitoring the quality of care

This final category, focuses on evaluating clinical outcomes through survival analysis. Using chronologically validated data such as: diagnosis dates, treatment initiation, and progression or death events the system supports the computation of key outcome metrics, including Overall Survival (OS) and Progression-Free Survival (PFS). These analyses enable clinicians to monitor treatment effectiveness and assess the quality of care provided. The results are presented through a dashboard that includes survival curves and cohort level summaries. Addition-

ally, the system supports cohort selection and the exportation of results in formats compatible with external statistical tools or universally readable formats such as PDF and CSV.

Everything described above can be seen as requirements on the below table 5.3, for further specification refer to appendixes Monitoring the Quality of Care and Oyster Level Requirements.

| ID    | Requirement                 | Description                                                                                                                                                                                                                                                                        |
|-------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR2.1 | Retrieve treatment data     | The system must retrieve patient treatment records, including: Diagnosis date; First cycle date; Progression date / End of treatment; Death date; Status (0/1): 0: censored (patient did not have the event until the end of the study); 1: event occurred (progression or death). |
| FR2.2 | Compute OS and PFS          | The system must calculate Overall Survival (OS) and Progression-Free Survival (PFS).                                                                                                                                                                                               |
| FR2.3 | Apply Kaplan-Meier analysis | The system must use Kaplan-Meier survival estimation to analyze outcomes.                                                                                                                                                                                                          |
| FR2.4 | Visualize results           | The system must generate survival curves and statistical summaries.                                                                                                                                                                                                                |

Table 5.3: Monitoring the Quality of Care – Functional Requirements

# Chapter 6

## Architecture

In this chapter there will be a clear description of the proposed system architecture, following the C4 Model. The objective of this architectural description is to provide a structured and progressive view of the system, from its external interactions to its internal organization. At the current stage of development, only the first three abstraction levels of the C4 Model were designed, namely the Context, Container, and Component levels. This architecture should not be interpreted as final and may be subject to refinements and changes throughout the course of the second semester, as implementation and experimental validation progress.

### 6.1 Context

At this level, the context diagram presents the system as a black box and describes how it interacts with users and external systems, [Vazquez-Ingelmo et al., 2020]. The objective of this diagram is to clearly define the system boundaries and to identify which actors and external dependencies are involved in its operation.

The primary user of the system is defined as the clinician, who directly interacts with the system in order to perform patient screening, clinical trial analysis, and monitoring of care. The system itself, represented as CTESS, aggregates all functionalities previously described, including structured data extraction, patient-trial matching, and survival analysis.

CTESS communicates with a single external system, SClinico, which is the platform used by ULS Coimbra for the storage of free-text clinical diaries. This interaction is limited to data acquisition and has the objective of retrieving unstructured clinical records that will later be processed and transformed by the system. No direct modifications are performed on SClinico, ensuring that the proposed system remains non-intrusive with respect to existing clinical infrastructures.

Everything discussed above is present in Figure 6.1.

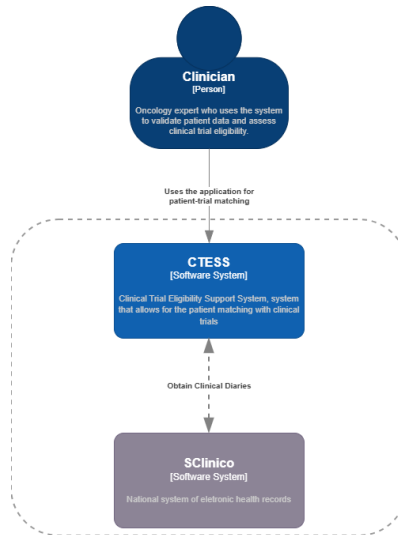


Figure 6.1: Context Diagram of the system

## 6.2 Container

At this abstraction level, the system is decomposed into its main runtime units, each one having a well-defined responsibility and interaction pattern, [Vazquez-Ingelmo et al., 2020].

- **Core Application**, which represents the central backend of the system. It exposes the main API consumed by the user interface and orchestrates all system functionalities, including data extraction, patient-trial matching, and monitoring of care.
- **Web Application**, responsible for delivering the static user interface assets. It presents the extracted parameters, logical rules, eligibility reports, and monitoring results to the user in a structured and readable format.
- **Single-Page Application**, which runs in the user's web browser and provides the interactive functionality of the system. This component acts as a bridge between the Web Application and the Core Application, handling user input and API communication.
- **Database**, which stores all relevant system data. This includes unstructured clinical diaries, structured patient profiles, extracted and validated clinical trial criteria, eligibility results, and monitoring outcomes.

Everything discussed above is represented in Figure 6.2, which illustrates the separation of concerns and the communication paths between the different containers.

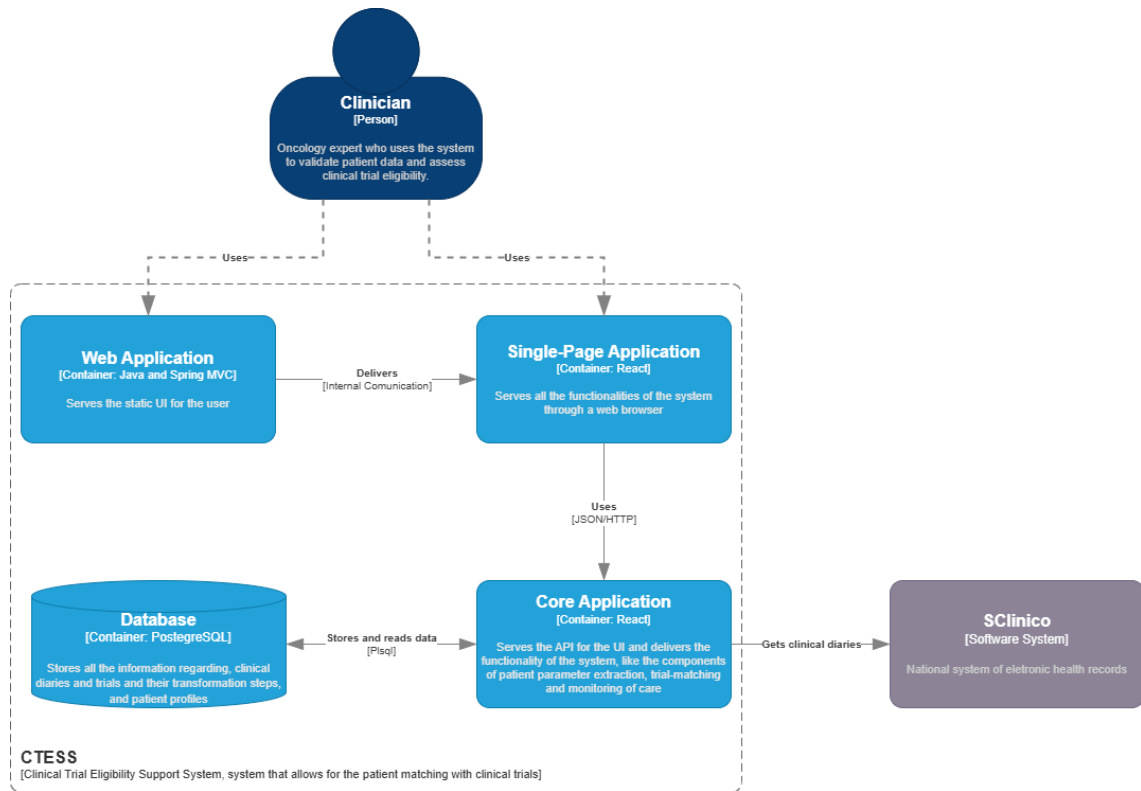


Figure 6.2: Container Diagram of the system

### 6.3 Component

This level focuses on the internal organization of the Core Application, identifying the main components that implement the system's functional requirements, [Vazquez-Ingelmo et al., 2020]. It is composed of:

- Main component, which is responsible for centralizing and orchestrating all other components. It exposes the API consumed by the user interface and manages the overall execution flow of the system.
- Patient Parameter Extraction Component, which encapsulates all functionalities related to the Automated Structured Data Extraction module. This component is responsible for processing unstructured clinical diaries, extracting relevant parameters, and storing the resulting structured data.
- Patient-Trial Matching Component, which implements the Identification of Eligible Patients for Clinical Trials module. It handles the extraction and validation of clinical trial criteria, their conversion into logical rules, and the deterministic evaluation of patient eligibility.
- Monitoring of Care Component, which corresponds to the Monitoring the quality of care module. This component retrieves treatment data and performs survival analysis, including the computation and visualization of metrics such as Overall Survival and Progression-Free Survival.

Everything discussed above is represented in Figure 6.3, which provides a detailed view of how responsibilities are distributed within the Core Application.

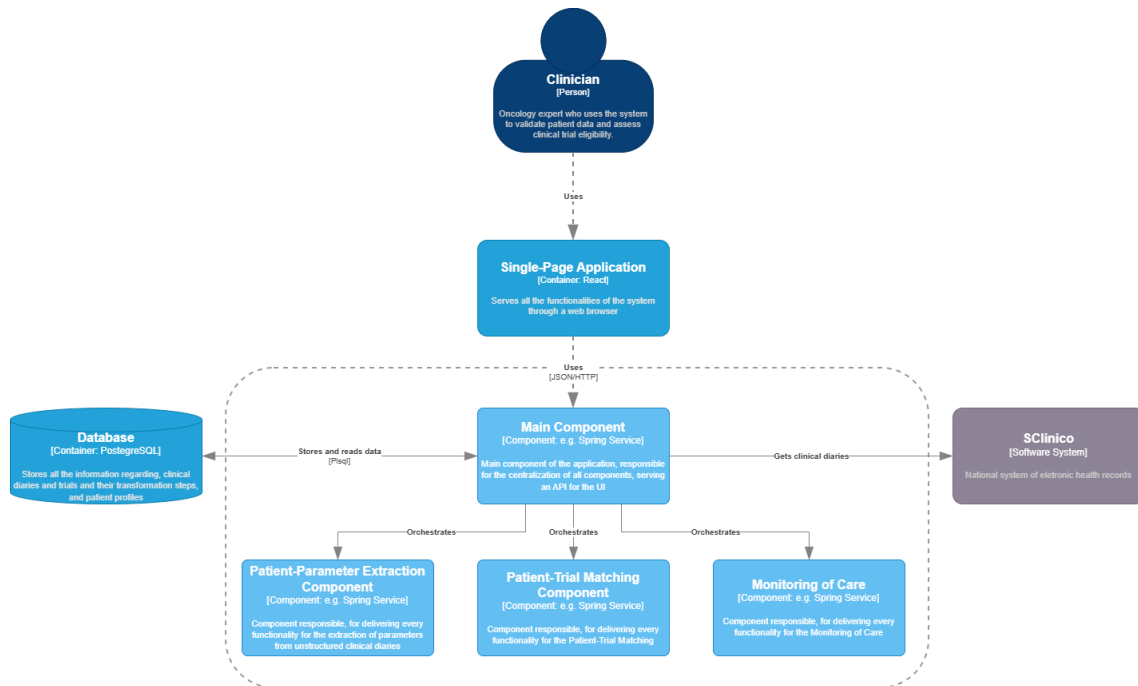


Figure 6.3: Component Diagram of the system

# Chapter 7

## Experimental Setup

In this section, the experimental setup for the upcoming semester is formally defined. Its purpose is to guide the exploratory analysis and ensure methodological consistency throughout the study. The following subsections outline the main objectives, the dataset selected for experimentation, the LLMs to be evaluated, the metrics adopted for assessment, and the scenarios under which the experiments will be conducted.

### 7.1 Experimental Objectives

As discussed previously, the proposed system comprises two main LLM-based pipelines:

- Automated structured data extraction;
- Conversion of clinical trial criteria into logical rules.

Accordingly, the experiments to be conducted aim to evaluate the accuracy and robustness of the LLM-based components employed in these pipelines, as well as their suitability for integration into a semi-automated, clinician-in-the-loop system. Particular emphasis is placed on assessing whether the proposed approach can reliably support clinical workflows while preserving transparency and expert control.

### 7.2 Dataset and Data Preparation

The experimental setup relies on two primary data sources:

- Unstructured clinical diaries, containing free-text oncology patient records;
- Clinical trial documents, describing in natural language the inclusion and exclusion criteria.

The source of the clinical diaries is *SClinico*, which contains all available oncology clinical diary data. These documents are stored in a PostgreSQL database as raw text, following the procedure described in FR0.1 - Load historical data. Clinical trial documents are provided either as plain text extracted from publicly available trial descriptions or as documents supplied directly by the project stakeholders.

No manual correction or pre-processing is applied prior to LLM processing, ensuring that the evaluation reflects realistic clinical conditions and the inherent variability of real-world clinical documentation.

### 7.3 LLMs Evaluated

The experiments evaluate multiple Large Language Models by analyzing their performance across the proposed tasks. The selected models represent different model families and architectural approaches:

- Qwen3 8B Instruct;
- Mistral Small 3.1 Instruct;
- Llama 3.1 8B Instruct.

All models are accessed using deterministic inference settings to minimize variability and ensure reproducibility. No fine-tuning is performed, and all experiments are conducted in a zero-shot configuration, in line with the intended deployment constraints of the system.

### 7.4 Prompting and Inference Configuration

All LLM interactions follow fixed and standardized prompt templates, differing only in the injected clinical content and the specific task being executed. This approach ensures that comparisons between models are conducted under fair and controlled conditions. Examples of these prompt templates are presented in FR3.2 - Extract structured info and FR3.3 - Normalize text.

These prompts explicitly define the role of the LLM, the task to be performed, the expected output structure, and constraints on the output format and allowed values.

Inference parameters such as temperature are set to zero (or the minimum allowed value) to reduce stochasticity. Each document is processed independently, ensuring traceability and auditability at the individual document level.

## 7.5 Evaluation Metrics

The evaluation employs both quantitative and qualitative metrics, depending on the task under analysis.

For structured data extraction, the following metrics are used: field-level accuracy (correct versus incorrect extraction), missing field rate (fields returned as None), and schema compliance rate.

For eligibility criteria conversion, the following metrics are used: correct classification of inclusion and exclusion criteria, and logical correctness of the generated rules, validated by a domain expert and graded as correct or incorrect.

Finally, for eligibility decision justification, the following metrics are used: clinical coherence of the generated explanations, graded by experts on a scale from 0 to 5, and consistency between the deterministic eligibility decision and the generated justification, classified as consistent, slightly inconsistent, or completely inconsistent.

## 7.6 Experimental Scenarios

The experiments are organized into the following scenarios:

1. Single-diary extraction: LLMs are evaluated and compared on the task of extraction accuracy for individual clinical records;
2. Single-trial matching: LLMs are evaluated and compared on the task of converting clinical trial eligibility criteria into logical representations.

All comparisons between outputs generated by different LLMs are performed under identical conditions to ensure a reliable, fair, and robust experimental evaluation, allowing for a meaningful selection of the most suitable model.

To further ensure reproducibility, all prompts are versioned and stored, and all LLM configuration parameters are fixed to consistent and fair values across experiments.

An expert validation step is included for both extracted patient data and trial criteria conversion. This validation does not influence the LLM outputs but serves as a ground-truth reference for evaluation and error analysis.

# Chapter 8

## Conclusion

This thesis presented the framing, motivation, and main objectives for the development of a clinical decision-support system for the oncology services of ULS Coimbra.

The work focuses on transforming unstructured clinical data into structured patient profiles and systematically comparing them with eligibility criteria extracted and converted into logical rules from clinical trials. With the main goal of reducing the manual workload present in the current workflow and improving the efficiency of the patient-selection process for clinical trials.

In the next semester, the main focus will be the implementation of the system. This phase will include the final validation of requirements, experimentation with LLMs for information extraction and criteria conversion, development of the backend and user interface, integration of all components, and the execution of complete tests with realistic data. In parallel, the formalization of the thesis document will be concluded, accompanying the entire development process.

The work conducted during this semester progressed smoothly, with few issues. The difficulties that did arise were mostly related to other curricular units, which occasionally diverted attention from the thesis and project to other academic responsibilities. Although this may have delayed some aspects of the first-semester work, it had no significant impact on the overall progress. Considering the objectives defined for this phase, the work carried out can be viewed as successful.

# References

- Proprietary Model | LLM Knowledge Base. URL <https://promptmetheus.com/resources/llm-knowledge-base/proprietary-model>.
- Open-Source Models | Definition, Benefits & Key Challenges. URL <https://xenoss.io/ai-and-data-glossary/open-source-models>.
- Open Model: Accessible Algorithms & Collaborative Development, January 2025. URL <https://www.lyzr.ai/glossaries/open-model/>.
- Navya Bhagat and Olivia Mackey. Large Language Models for Efficient Medical Information Extraction.
- Banghao Chen, Zhaofeng Zhang, Nicolas Langrené, and Shengxin Zhu. Unleashing the potential of prompt engineering for large language models. *Patterns*, 6(6):101260, June 2025. ISSN 26663899. doi: 10.1016/j.patter.2025.101260. URL <https://linkinghub.elsevier.com/retrieve/pii/S2666389925001084>.
- Shilin Ding, Gao Cong, Chin-Yew Lin, and Xiaoyan Zhu. Using Conditional Random Fields to Extract Contexts and Answers of Questions from Online Forums.
- Michael R. Douglas. Large Language Models, October 2023. URL <http://arxiv.org/abs/2307.05782>. arXiv:2307.05782 [cs].
- Bowen Gu, Vivian Shao, Ziqian Liao, Valentina Carducci, Santiago Romero Brufau, Jie Yang, and Rishi J. Desai. Scalable information extraction from free text electronic health records using large language models. *BMC Medical Research Methodology*, 25(1):23, January 2025. ISSN 1471-2288. doi: 10.1186/s12874-025-02470-z. URL <https://bmcmmedresmethodol.biomedcentral.com/articles/10.1186/s12874-025-02470-z>.
- Shashi Gupta, Aditya Basu, Mauro Nievas, Jerrin Thomas, Nathan Wolfrath, Adhitya Ramamurthi, Bradley Taylor, Anai N. Kothari, Regina Schwind, Therica M. Miller, Sorena Nadaf-Rahrov, Yanshan Wang, and Hrituraj Singh. PRISM: Patient Records Interpretation for Semantic clinical trial Matching system using large language models. *npj Digital Medicine*, 7(1):305, October 2024. ISSN 2398-6352. doi: 10.1038/s41746-024-01274-7. URL <https://www.nature.com/articles/s41746-024-01274-7>.
- Michael Hahn and Navin Goyal. A Theory of Emergent In-Context Learning as Implicit Structure Induction, March 2023. URL <http://arxiv.org/abs/2303.07971>. arXiv:2303.07971 [cs].

## REFERENCES

---

- Aaron Hooper. Prompt Engineering and In-Context Learning: A Comprehensive Survey of Techniques, Applications, and Future Directions.
- Qiao Jin, Zifeng Wang, Charalampos S. Floudas, Fangyuan Chen, Changlin Gong, Dara Bracken-Clarke, Elisabetta Xue, Yifan Yang, Jimeng Sun, and Zhiyong Lu. Matching patients to clinical trials with large language models. *Nature Communications*, 15(1):9074, November 2024. ISSN 2041-1723. doi: 10.1038/s41467-024-53081-z. URL <https://www.nature.com/articles/s41467-024-53081-z>.
- Liteng Liu, Sen Zhang, Yangmin Jiang, Jingzhen Guo, and Wenlong Zhao. Task Decomposition and Self-Evaluation Mechanisms for Home Healthcare Robots Using Large Language Models. *IEEE Access*, 13:65726–65736, 2025. ISSN 2169-3536. doi: 10.1109/ACCESS.2025.3559076. URL <https://ieeexplore.ieee.org/document/10960333/>.
- Mauro Nieves, Aditya Basu, Yanshan Wang, and Hrituraj Singh. Distilling Large Language Models for Matching Patients to Clinical Trials, December 2023. URL <http://arxiv.org/abs/2312.09958>. arXiv:2312.09958 [cs].
- Vasileios Ntinopoulos, Hector Rodriguez Cetina Biefer, Igor Tudorache, Nestoras Papadopoulos, Dragan Odavic, Petar Risteski, Achim Haeussler, and Omer Dzemali. Large language models for data extraction from unstructured and semi-structured electronic health records: a multiple model performance evaluation. *BMJ Health & Care Informatics*, 32(1):e101139, January 2025. ISSN 2632-1009. doi: 10.1136/bmjhci-2024-101139. URL <https://informatics.bmj.com/lookup/doi/10.1136/bmjhci-2024-101139>.
- Miles Osborne. Using maximum entropy for sentence extraction. In *Proceedings of the ACL-02 Workshop on Automatic Summarization -*, volume 4, pages 1–8, Philadelphina, Pennsylvania, 2002. Association for Computational Linguistics. doi: 10.3115/1118162.1118163. URL <http://portal.acm.org/citation.cfm?doid=1118162.1118163>.
- Mojdeh Rahmanian, Seyed Mostafa Fakhrahmad, and Seyedeh Zahra Mousavi. Towards Efficient Patient Recruitment for Clinical Trials: Application of a Prompt-Based Learning Model.
- Maciej Rybinski, Wojciech Kusa, Sarvnaz Karimi, and Allan Hanbury. Learning to match patients to clinical trials using large language models. *Journal of Biomedical Informatics*, 159:104734, November 2024. ISSN 15320464. doi: 10.1016/j.jbi.2024.104734. URL <https://linkinghub.elsevier.com/retrieve/pii/S1532046424001527>.
- Andrei N. Soklakov. Occam’s Razor as a Formal Basis for a Physical Theory, February 2002. URL <http://arxiv.org/abs/math-ph/0009007>. arXiv:math-ph/0009007.
- Minjun Son, Yun-Jae Won, and Sungjin Lee. Optimizing Large Language Models: A Deep Dive into Effective Prompt Engineering Techniques. *Applied Sciences*, 15(3):1430, January 2025. ISSN 2076-3417. doi: 10.3390/app15031430. URL <https://www.mdpi.com/2076-3417/15/3/1430>.

- Xuanming Su. Hidden Markov Models for Information Extraction.
- Madhumita Sushil, Vanessa E. Kennedy, Divneet Mandair, Brenda Y. Miao, Travis Zack, and Atul J. Butte. CORAL: Expert-Curated Oncology Reports to Advance Language Model Inference. *NEJM AI*, 1(4), March 2024. ISSN 2836-9386. doi: 10.1056/AIdbp2300110. URL <https://ai.nejm.org/doi/10.1056/AIdbp2300110>.
- Andrea Vazquez-Ingelmo, Alicia Garcia-Holgado, and Francisco J. Garcia-Penalvo. C4 model in a Software Engineering subject to ease the comprehension of UML and the software. In *2020 IEEE Global Engineering Education Conference (EDUCON)*, pages 919–924, Porto, Portugal, April 2020. IEEE. ISBN 978-1-7281-0930-5. doi: 10.1109/EDUCON45650.2020.9125335. URL <https://ieeexplore.ieee.org/document/9125335/>.
- Dinithi Vithanage, Chao Deng, Lei Wang, Mengyang Yin, Mohammad Alkhalaf, Zhenyu Zhang, Yunshu Zhu, Alan Christy Soewargo, and Ping Yu. Evaluating approaches of training a generative large language model for multi-label classification of unstructured electronic health records, June 2024. URL <http://medrxiv.org/lookup/doi/10.1101/2024.06.24.24309441>.
- Dinithi Vithanage, Ping Yu, Qianqian Xie, Hua Xu, Lei Wang, and Chao Deng. A comprehensive evaluation of large language models for information extraction from unstructured electronic health records in residential aged care. *Computers in Biology and Medicine*, 197:111013, October 2025. ISSN 00104825. doi: 10.1016/j.combiomed.2025.111013. URL <https://linkinghub.elsevier.com/retrieve/pii/S0010482525013654>.
- Jason Wei, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, Maarten Bosma, Denny Zhou, Donald Metzler, Ed H. Chi, Tatsunori Hashimoto, Oriol Vinyals, Percy Liang, Jeff Dean, and William Fedus. Emergent Abilities of Large Language Models, October 2022. URL <http://arxiv.org/abs/2206.07682>. arXiv:2206.07682 [cs].
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, January 2023. URL <http://arxiv.org/abs/2201.11903>. arXiv:2201.11903 [cs].
- Michael Wornow, Alejandro Lozano, Dev Dash, Jenelle Jindal, Kenneth W. Mahaffey, and Nigam H. Shah. Zero-Shot Clinical Trial Patient Matching with LLMs, April 2024. URL <http://arxiv.org/abs/2402.05125>. arXiv:2402.05125 [cs].
- Derong Xu, Wei Chen, Wenjun Peng, Chao Zhang, Tong Xu, Xiangyu Zhao, Xian Wu, Yefeng Zheng, Yang Wang, and Enhong Chen. Large Language Models for Generative Information Extraction: A Survey, October 2024. URL <http://arxiv.org/abs/2312.17617>. arXiv:2312.17617 [cs].

## REFERENCES

---

- Zhe Yang, Yichang Zhang, Yudong Wang, Ziyao Xu, Junyang Lin, and Zhi-fang Sui. Confidence v.s. Critique: A Decomposition of Self-Correction Capability for LLMs, December 2024. URL <http://arxiv.org/abs/2412.19513>. arXiv:2412.19513 [cs].
- Yulin Zhou, Yiren Zhao, Ilia Shumailov, Robert Mullins, and Yarin Gal. Revisiting Automated Prompting: Are We Actually Doing Better?, June 2023. URL <http://arxiv.org/abs/2304.03609>. arXiv:2304.03609 [cs].
- Yuxiang Zhou, Jiazheng Li, Yanzheng Xiang, Hanqi Yan, Lin Gui, and Yulan He. The Mystery of In-Context Learning: A Comprehensive Survey on Interpretation and Analysis. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 14365–14378, Miami, Florida, USA, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.795. URL <https://aclanthology.org/2024.emnlp-main.795>.

# Appendices

# Appendix A

## Requirements

In the following chapter, the main requirements of the project at hand will be discussed. These requirements were documented and specified with the clear objective of providing guidance throughout the development of the current project. They were structured using an oyster–sea level analogy: the sea-level requirements represent the highest and broadest level of abstraction, while the oyster-level requirements correspond to the most detailed and system-specific needs.

The requirements were obtained through numerous meetings with the client, and further refined in discussions with the advisor, tutor, and Dra. Margarida. This collaborative process ensured that every requirement is feasible, realistic, and makes sense within the hospital environment, both from a technical and operational perspective.

These requirements can be divided into four main sections, each possibly containing subsections of functional, data, and interface requirements. Below are the four identified sections:

- System requirements, requirements that will allow the proper functioning and overall stability of the system;
- Identification of eligible patients for clinical trials, requirements that will allow the identification of eligible patients (already structured) and match them to a relevant clinical trial;
- Monitoring the quality of care, requirements that will allow the verification and assessment of the level of success that a clinical trial has on the patients involved;
- Automated structured data extraction, requirements that will allow the extraction of structured data from unstructured patient EHRs, storing them for further use.

## A.1 Sea Level Requirements

### A.1.1 System Requirements

| ID    | Requirement                    | Description                                                                                                                                                                                               |
|-------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR0.1 | Load historical data           | The clinical trials since (a year) should be loaded into the system in order for it to start working.                                                                                                     |
| FR0.2 | Pre-process clinical documents | The system must clean and normalize the clinical documentation before extraction, removing irrelevant headers, footers, formatting artifacts, and duplicated text to ensure consistent input for the LLM. |

Table A.1: System Requirements - Functional Requirements

## A.1.2 Identification of Eligible Patients for Clinical Trials

### Functional Requirements

| ID    | Requirement                                    | Description                                                                                                                                                              |
|-------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR1.1 | Extract clinical trial criteria                | The LLM must extract inclusion and exclusion criteria from a clinical trial.                                                                                             |
| FR1.2 | Convert criteria into logic                    | The extracted text must be converted into logical expressions (e.g., Age > 18).                                                                                          |
| FR1.3 | Validation of inclusion and exclusion criteria | After the LLM extracts the inclusion and exclusion criteria, a second phase must occur in which the doctor validates these results in order for patients to be selected. |
| FR1.4 | Compare criteria with database                 | The system must automatically evaluate patient data against the logical criteria of clinical trials.                                                                     |
| FR1.5 | Generate eligibility report                    | The system must output a list of eligible patients and provide justification for excluded patients.                                                                      |
| FR1.6 | Saving included patients                       | After a patient is considered eligible for a trial, they should be paired with it and this information must be persisted.                                                |

Table A.2: Identification of Eligible Patients for Clinical Trials – Functional Requirements

### Data Requirements

| ID    | Requirement            | Description                                                                                                                                                                    |
|-------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DR1.1 | Patient profile fields | The database must contain, for each patient, the following fields: Age, Diagnosis, Treatment, Stage, etc., as described in the "Automated Structured Data Extraction" section. |
| DR1.2 | Criteria mapping       | Each criterion must map to the corresponding patient attributes in the database.                                                                                               |

Table A.3: Identification of Eligible Patients for Clinical Trials - Data Requirements

**Interface Requirements**

| <b>ID</b> | <b>Requirement</b>                  | <b>Description</b>                                                                         |
|-----------|-------------------------------------|--------------------------------------------------------------------------------------------|
| IR1.1     | Clinical trial submission           | The clinician must be able to submit several clinical trials into the system.              |
| IR1.2     | Clinical trial selection            | The clinician must be able to select a clinical trial and view eligibility results.        |
| IR1.3     | Reporting interface                 | The system must display the justification report in a readable format (PDF/HTML).          |
| IR1.4     | Clinical trial validation interface | A page must be displayed, allowing the doctor to edit and validate the extracted criteria. |

Table A.4: Identification of Eligible Patients for Clinical Trials - Interface Requirements

### A.1.3 Monitoring the Quality of Care

#### Functional Requirements

| ID    | Requirement                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR2.1 | Retrieve treatment data     | <p>The system must retrieve patient treatment records, including:</p> <ul style="list-style-type: none"> <li>• Diagnosis date;</li> <li>• First cycle date;</li> <li>• Progression date / End of treatment;</li> <li>• Death date;</li> <li>• Status (0/1): <ul style="list-style-type: none"> <li>– 0: censored (patient did not have the event until the end of the study);</li> <li>– 1: event occurred (progression or death).</li> </ul> </li> </ul> |
| FR2.2 | Compute OS and PFS          | The system must calculate Overall Survival (OS) and Progression-Free Survival (PFS).                                                                                                                                                                                                                                                                                                                                                                      |
| FR2.3 | Apply Kaplan-Meier analysis | The system must use Kaplan-Meier survival estimation to analyze outcomes.                                                                                                                                                                                                                                                                                                                                                                                 |
| FR2.4 | Visualize results           | The system must generate survival curves and statistical summaries.                                                                                                                                                                                                                                                                                                                                                                                       |

Table A.5: Monitoring the Quality of Care – Functional Requirements

For the requirement FR2.1, the extracted fields from the "Automated Structured Data Extraction" phase are mapped with the necessary fields for calculating the Kaplan-Meier:

| Extracted field            | Necessary field  | Obs                                                 |
|----------------------------|------------------|-----------------------------------------------------|
| Date of first treatment    | First cycle date |                                                     |
| Diagnosis date             | Diagnosis date   |                                                     |
| Death date                 | Death date       | If it exists: OS event = 1; otherwise OS event = 0. |
| End date (from treatments) | End of treatment | For PFS, event = 0                                  |
| -                          | Progression date | For PFS event = 1                                   |

Table A.6: Mapping of Extracted Fields to Necessary Fields

### Data Requirements

| ID    | Requirement          | Description                                                                           |
|-------|----------------------|---------------------------------------------------------------------------------------|
| DR2.1 | Required data fields | Diagnosis date, first cycle date, progression/end date, death date, and status (0/1). |
| DR2.2 | Derived variables    | OS = (death date - diagnosis date);<br>PFS = (progression date - first cycle date).   |

Table A.7: Monitoring the Quality of Care - Data Requirements

### Interface Requirements

| ID    | Requirement        | Description                                                                    |
|-------|--------------------|--------------------------------------------------------------------------------|
| IR2.1 | Analysis dashboard | The clinician must be able to select patient cohorts and view survival graphs. |
| IR2.2 | Report export      | Results should be exportable to CSV, PDF, or be compatible with SPSS.          |

Table A.8: Monitoring the Quality of Care - Interface Requirements

## A.1.4 Automated Structured Data Extraction

### Functional Requirements

| ID    | Requirement                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR3.1 | Select diaries for extraction      | The system must allow one or multiple diaries to be extracted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| FR3.2 | Extract structured info            | <p>The LLM must extract patient data from free-text diaries, including:</p> <ul style="list-style-type: none"> <li>• Age / Date of birth;</li> <li>• ECOG-PS — describes a patient's level of functioning in terms of their ability to care for themselves, daily activity, and physical ability;</li> <li>• Diagnosis;</li> <li>• Date of diagnosis;</li> <li>• Molecular status;</li> <li>• Stage;</li> <li>• Date of treatment;</li> <li>• Treatments (First, Second);</li> <li>• Start date;</li> <li>• End date;</li> <li>• Control.</li> </ul> <p>If a field does not exist in the clinical diary, it must be defined as None.</p> |
| FR3.3 | Normalize text                     | The system must correct typographical and linguistic inconsistencies.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| FR3.4 | Validation of extracted parameters | After the LLM extracts the parameters, the doctor can edit and validate these data.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| FR3.5 | Store structured profiles          | Each structured patient record must be stored in the local database.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

Table A.9: Automated Structured Data Extraction - Functional Requirements

## Data Requirements

| ID    | Requirement      | Description                                                                                                                                                       |
|-------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DR3.1 | Extracted fields | Age / date of birth, ECOG-PS, Diagnosis, Date of Diagnosis, Stage, Molecular Status, Date of Treatment, Treatments, Treatment start date, and Treatment end date. |
| DR3.2 | Storage schema   | All extracted data must be stored in a PostgreSQL database.                                                                                                       |

Table A.10: Automated Structured Data Extraction - Data Requirements

## A.2 Oyster Level Requirements

### A.2.1 FR0.1 - Load historical data

All available unstructured clinical diary entries on SClinico for the oncology must be loaded into the PostgreSQL Database.

This will be done through a full load python script where unstructured data is read and stored in the a dedicated table on the DB. Because of the sheer amount of data it is unfeasible for it to be a manual process.

### A.2.2 FR0.2 - Pre-process clinical documents

Before any extraction or matching can occur, all clinical documents must undergo a pre-processing phase to ensure that the input provided to the LLM is clean, consistent, and free from structural noise. This step includes the removal of irrelevant headers, footers, duplicated sections, and formatting artifacts commonly present in exported clinical diaries or trial documentation. The pre-processing will be executed automatically as part of the ingestion pipeline. Each document will be parsed, normalized, and transformed into a standardized text format, ensuring that downstream extraction processes operate on uniform and reliable input.

In addition, the pre-processing logic must be tailored to the specific type of document being ingested.

- For clinical trials, the system must isolate and retain only the inclusion and exclusion criteria sections, discarding unrelated protocol content such as study rationale, endpoints, administrative metadata, or regulatory notes.
- For clinical diaries, the system must preserve only the textual content relevant to patient information and remove any extraneous elements, ensuring that the LLM receives clean input for extracting the required clinical parameters: demographic information, ECOG-PS, primary diagnosis, date of

diagnosis, molecular status, cancer stage, treatment history , and control information.

### A.2.3 FR1.1 - Extract clinical trial criteria

The LLM must extract inclusion and exclusion criteria from a clinical trial. This process must:

- Parse the unstructured text of the clinical trial
- Identify all eligibility criteria
  - Inclusion Criteria - under the "Inclusion Criteria" section on the clinical trial
  - Exclusion Criteria - under the "Exclusion Criteria" section on the clinical trial
  - If by any chances these sections aren't present the LLM must be capable of classifying the eligibility criteria into the respective categories.

### A.2.4 FR1.2 - Convert Criteria into logic

After extracting inclusion and exclusion criteria from the clinical trial text (FR1.1), the system must use the LLM to convert each criterion into a structured, machine-readable logical representation. This process must:

- Transform free-text criteria into logical rules (e.g. comparison operators, Boolean conditions)
- This step must output the result in a standardized JSON logic format:

```
{
  "criterion": {
    "type": "inclusion/exclusion",
    "logic": "[FIELD][OPERATOR][VALUE]"
  }
}
```

- This conversion must be stored in the PostgreSQL table `clinical_trial_criteria` for further use.

### A.2.5 FR1.3 - Validation of inclusion and exclusion criteria

After the extraction and conversion of criteria of inclusion and exclusion (FR1.2 - Convert Criteria into logic) a second phase must be undergone, the doctor must validate this results in order for the patients to be elected. This must be done by:

- Displaying all extracted criteria in a dedicated interface where the doctor can:
  - Review each inclusion and exclusion criterion;
  - Edit incorrect or ambiguous text extracted from the clinical trial
  - Modify or correct the associated logical rule (operators, field and values).
- The system must only run the patient-trial matching (FR1.4 - Compare criteria with database) when all criteria for that trial are validated;
- All clinician modifications must be persisted in PostgreSQL.

### A.2.6 FR1.4 - Compare criteria with database

For a given clinical trial, the system must recover and evaluate patient structured data against the logical criteria of clinical trials produced in FR1.2 - Convert Criteria into logic. This process must:

- Retrieve all structured patient records from the PostgreSQL database;
- Retrieve all logical eligibility criteria associated with the selected clinical trial from the `trial_criteria_logic` table;
- Deterministically evaluate each patient against every inclusion and exclusion rule:
  - Inclusion criteria must all be satisfied;
  - Exclusion criteria must not be satisfied;
  - The evaluation must not rely on randomness or LLM inference at this stage.
- Generate an eligibility decision for each patient:
  - 1 - patient satisfies all inclusion criteria and violates none of the exclusion criteria;
  - 0 - otherwise

### A.2.7 FR1.5 - Generate eligibility report

After the result of FR1.4 - Compare criteria with database. the system must output a list of eligible/excluded patients and LLM justification for each decision. The process must be done by:

- Request a justification for the decision from an LLM, having in account:
  - Inclusion and exclusion criteria;

- Structured fields of the patient;
- Deterministic decision result.
- Produce a structured dictionary combining the id and name of the patient, previous eligibility decision and the justification produced:

```
eligible_list = {  
    "Patient" : f"{patient_id}-{Name}",  
    "Decision" : 0/1,  
    "Justification" : "<LLM_prompt>"  
}
```

### A.2.8 FR1.6 - Saving included patients

After a patient is considered eligible for a trial, he should be paired with it and this information must be persisted. So it should:

- Store the final eligibility results (decision + justification) in the PostgreSQL table `patient_trial_eligibility` this table connects the patients with the trial.
- Both included and excluded patients must have their decision and justification saved on it for audit purposes.

### A.2.9 FR2.1 - Retrieve treatment data

The system must retrieve all patient treatment-related records required for survival analysis. This process must:

- Load, for each patient, the following structured fields from the PostgreSQL database (previously extracted and validated in the "Automated Structured Data Extraction" phase):
  - Diagnosis date;
  - Date of first treatment (mapped to First cycle date);
  - Progression date (if present);
  - End date of treatment (if no progression occurred);
  - Death date (if applicable).
- Automatically derive the Status (0/1) required for Kaplan-Meier calculations based on the available data:
  - OS status rules:
    - \* 1 - if a death date exists;
    - \* 0 - otherwise (censored).

- PFS status rules
  - \* 1 - if a progression date exists;
  - \* 0 - if progression did not occur and only end-of-treatment is available (censored)
- Apply the following deterministic mapping rules between extracted fields on the "Automated Structured Data Extraction" phase and Kaplan-Meier required fields:

| Extracted field            | Required KM field | Notes                                    |
|----------------------------|-------------------|------------------------------------------|
| Date of first treatment    | First cycle date  | Direct mapping                           |
| Diagnosis date             | Diagnosis date    | Direct mapping                           |
| Death date                 | Death date        | OS event = 1 if present, else event = 0. |
| End date (from treatments) | End of treatment  | Used when no progression; PFS event = 0  |
| -                          | Progression date  | if present then PFS event = 1            |

Table A.11: Mapping of Extracted Fields to Necessary Fields

### A.2.10 FR2.2 - Compute OS and PFS

After retrieving and mapping all necessary fields in FR2.1, the system must compute the survival metrics Overall Survival (OS) and Progression-Free Survival (PFS) for each patient. This process must:

- Use standardized date formats;
- Apply the following logic for Overall Survival (OS):
  - If death date exists:
    - \*  $OS = (\text{death date} - \text{diagnosis date})$
- Apply the following logic for Progression-Free Survival (PFS):
  - If progression date exists:
    - \*  $PFS = (\text{progression date} - \text{first cycle date})$

### A.2.11 FR2.3 - Apply Kaplan-Meier analysis

After computing OS and PFS for each patient (FR2.2), the system must generate survival estimations using the Kaplan-Meier method. This process must:

- Compute the Kaplan–Meier survival function deterministically using the standard formula:

$$S(t) = \prod_{t_i < t} \left(1 - \frac{d_i}{n_i}\right)$$

Where:

- $S(t)$ , probability of surviving beyond time  $t$ ;
  - $t_i$ , specific time points at which an event occurs;
  - $d_i$ , number of events that occur at time  $t_i$ ;
  - $n_i$ , number of individuals who are still at risk of the event just before time  $t_i$ .
- Ensure implementation includes:
    - Grouping of patients by survival time to compute  $d_i$  and  $n_i$ ;
    - Proper handling of censored observations (included in the risk set but not counted as events);
    - Stepwise survival curves (piecewise constant function).
  - The Kaplan-Meier estimates must be outputted into a structured format suitable for visualization (FR2.4 - Visualize results).
    - Time points;
    - Survival probabilities;
    - Number at risk;
    - Number of events;
    - Censoring information.

### A.2.12 FR2.4 - Visualize results

After computing the Kaplan–Meier survival estimates (FR2.3 - Apply Kaplan-Meier analysis), the system must generate interactive and exportable visualizations for clinician analysis. This process must:

- Produce visualization-ready survival curves with the following characteristics:
  - Stepwise Kaplan-Meier survival function;
  - Distinct curves for OS and PFS;
  - Markers for censored patients (e.g., "+" symbols);
  - Time displayed on the x-axis and survival probability on the y-axis.
- Allow clinicians to:
  - Select the patient cohort or subgroup to be visualized

- Toggle between OS and PFS views
  - Overlay multiple curves (e.g., by treatment group or stage)
  - Display underlying survival tables (number at risk per time interval)
- Ensure the graphical output can be:
  - Rendered interactively in the system's dashboard;
  - Exported in standard formats (PNG/SVG for images, PDF for reports);
  - Accompanied by summary statistics (median survival, events count, censored count).

### **A.2.13 FR3.1 - Select Diaries for extraction**

After the full load of unstructured clinical diaries into the PostgreSQL database (FR0.1 - Load historical data), the system must allow the clinician to select on or multiple diaries to undergo structured data extraction. This process must:

- Retrieve and display all available unstructured diary entries stored in the database.
- Provide an interface allowing the clinician to:
  - Select individual diaries manually;
  - Select multiple diaries using batch selection.
- Validate the selection to ensure:
  - Only diaries that have not yet undergone structured extraction (??) can be selected;
  - Duplicate extractions are prevented;
  - Any missing or corrupted diaries trigger a system warning.
- Forward selected diaries to the structure extraction pipeline (FR3.2 - Extract structured info-FR3.4 - Validation of extracted parameters) ensuring the system:
  - Records the diary status to "In Extraction" on the database.

### **A.2.14 FR3.2 - Extract structured info**

After the selection of relevant unstructured clinical diaries (FR3.1 - Select Diaries for extraction), the system must use an LLM-based extraction pipeline to convert free-text diary content into structured patient data. The process must:

- Process each selected diary independently to ensure reproducibility and traceability;

- Each clinical diary must be injected into the LLM through a deterministic zero-shot prompting pipeline, implemented via a Python script;
- For every diary, the script must construct a standardized prompt template, replacing only the free-text diary content to ensure consistent outputs;
- The prompt must include:
  - Clear extraction instructions
  - A fixed schema the model must respect
  - The raw diary text, inserted verbatim in a designated section, avoiding formatting that may leak metadata or context (e.g. <Diary> <\Diary>)
- The full prompt template should be structured like this:

```
You are a clinical information extraction assistant.
Your task is to extract structured clinical data from
an unstructured free text clinical diary.
```

```
Follow these rules:
```

- Extract only the information explicitly present in the diary.
- If a field is missing, set its value to null.
- Do not infer or guess clinical data.
- Output only valid JSON, respecting the specified schema.

```
Schema (all fields must appear in the output, null if
missing):
```

```
{
  "age_or_birthdate": ...,
  "ecog_ps": ...,
  "diagnosis": ...,
  "diagnosis_date": ...,
  "molecular_status": ...,
  "stage": ...,
  "treatments": {
    "1_treatment": {
      "start_date": ...,
      "end_date": ...
    },
    "2_treatment": {
      "start_date": ...,
      "end_date": ...
    }
  },
  "control": ...
}
```

```
Clinical diary to process:
<DIARY>
```

```
{{DIARY_TEXT}}
</DIARY>
```

Return only the JSON object.

- Extract the following clinical parameters from unstructured text:
  - Demographic information: Age and/or Date of birth.
  - Clinical performance metrics: ECOG-PS (Eastern Cooperative Oncology Group - Performance Status).
  - Diagnosis-related information: Primary diagnosis; Date of diagnosis; Molecular status; Cancer stage (TNM or equivalent).
  - Treatment information: Date of first recorded treatment; Treatments, for each: Start and End date.
  - Control information.
- Ensure robustness of extraction by:
  - Handling missing information gracefully, if a field is not present in the diary, the system must assign the value null;
  - Logging extraction confidence scores or flags for ambiguous fields.
- Produce a structured JSON object for each processed diary in a standardized schema compliant with the database model;
- Forward the extracted structured data to the next steps: Normalize text (FR3.3 - Normalize text); Validation of extracted parameters (FR3.4 - Validation of extracted parameters)

### A.2.15 FR3.3 - Normalize text

The system must correct typographical and linguistic inconsistencies in the extracted clinical fields. This normalization step is executed after the first LLM completes the extraction (FR3.2 - Extract structured info), and it must be performed individually for each extracted diary to ensure data integrity and traceability. The normalization process must:

- Receive as input the structured JSON generated by the previous extraction stage;
- Apply a second deterministic LLM prompt, using a fixed template, to normalize only the textual fields, while preserving numerical and date fields unchanged;
- Ensure consistency in spelling, abbreviations, medical terminology, and capitalization;
- Prohibit inference, or correction beyond language normalization;

- Ensure that missing fields remain null, without introducing new data;
- Produce a normalized JSON, following the exact same schema as in FR3.2 - Extract structured info, overwriting only the corrected text fields;
- The full normalization prompt template should be structured like this:

```
You are a clinical text normalization assistant.
Your task is to correct typographical and linguistic
inconsistencies in the extracted patient data.
```

```
Follow these rules:
```

- Only correct language/format inconsistencies (spelling, abbreviations, capitalization, terminology).
- Do NOT infer missing data or invent information.
- Keep numeric, date, and boolean fields unchanged.
- Output the result strictly in valid JSON, following the schema provided.

```
Schema (all fields must appear in the output, null if
missing):
```

```
{
  "age_or_birthdate": ...,
  "ecog_ps": ...,
  "diagnosis": ...,
  "diagnosis_date": ...,
  "molecular_status": ...,
  "stage": ...,
  "treatments": {
    "1_treatment": {
      "start_date": ...,
      "end_date": ...
    },
    "2_treatment": {
      "start_date": ...,
      "end_date": ...
    }
  },
  "control": ...
}
```

```
Here is the extracted JSON from the diary to normalize
:
```

```
<EXTRACTED_JSON>
{{EXTRACTED_JSON}}
</EXTRACTED_JSON>
```

```
Return only the corrected JSON object.
```

### A.2.16 FR3.4 - Validation of extracted parameters

After the LLM completes the extraction (FR3.2 - Extract structured info) and normalization (FR3.3 - Normalize text) of patient parameters, the system must provide a clinician validation interface to review, edit, and approve the structured data before it is persisted for further analysis. This process must:

- Retrieve the normalized structured JSON for each selected diary from the database;
- Display all extracted fields in a interface, organized by: Demographics (Age, date of birth); Performance status (ECOG-PS); Diagnosis information (diagnosis, date of diagnosis, stage, molecular status); Treatments (start date, end date); Control.
- Allow clinicians to:
  - Edit any incorrectly extracted or normalized field;
  - Add missing data if the diary contains information that was not captured
  - Delete erroneously extracted fields or treatments
  - Approve the validated dataset once all required fields are reviewed
- Ensure that validation status is tracked:
  - Each diary receives a validated = TRUE/FALSE flag
  - Record the timestamp for each validation session
  - Store any clinicians edits as a versioned log to ensure auditability
- Only after validation, the structured data should be used for downstream processes: Patient eligibility evaluation for clinical trials (FFR1.3 - Validation of inclusion and exclusion criteria); Survival analysis (FR2.1 - Retrieve treatment data-FR2.4 - Visualize results).
- Provide reproducibility and traceability by: Logging the original extracted JSON, normalized JSON, and the final validated JSON, maintaining a history of changes for auditing and quality control.

### A.2.17 FR3.5 - Store structured profiles

After extraction (FR3.2 - Extract structured info), normalization (FR3.3 - Normalize text), and clinician validation (FR3.4 - Validation of extracted parameters), the system must persist each structured patient record in the local PostgreSQL database. This process must:

- Store all validated patient records in a dedicated table `patient_profiles` with a schema that includes: Patient identifier; Demographics (Age/Date of

birth); Performance status (ECOG-PS); Diagnosis details (diagnosis, date of diagnosis, stage, molecular status); Treatments (start and end dates); Control.

- All changes to the unstructured diary, extraction, normalization and validation, must also be persisted in a versioning table `diary_log` as well as the timestamps for these;
- In the `patient_profiles` table it is necessary to ensure that: Only clinician-validated data (FR3.4 - Validation of extracted parameters) is stored; Fields not present in the diary are stored as NULL; All date fields are stored in standardized ISO format.
- The unstructured Diary must have its status updated on the database to "Validated" on the database.

# Appendix B

## Abstract Templates

### B.0.1 Clinical Trial Template

Listing B.1: Canonical structure of a clinical trial protocol

#### Study Objectives

- Primary objective:  
Description of the main clinical question addressed by the trial
- Secondary objectives:  
Description of additional clinical outcomes of interest (e.g., quality of life, functional status, safety profiles).
- Exploratory objectives:  
Optional objectives such as biomarker discovery or subgroup analysis

#### Study Design

- Study phase (e.g., Phase II, Phase III)
- Study type (e.g., randomized, open-label, double-blind)
- Allocation ratio (e.g., 1:1, 2:1)
- Control arm and experimental arm description
- Stratification factors (e.g., ECOG status, histology)

#### Number of Patients

- Planned total sample size
- Allocation per study arm
- Expected number of screened patients (if applicable)

#### Number of Sites

- Number of participating clinical centers
- Geographic distribution (e.g., countries or regions)

#### Eligibility Criteria

##### Inclusion Criteria:

- Demographic requirements (e.g., age, sex)
- Disease characteristics (e.g., diagnosis, stage)

- Molecular or biomarker status (if applicable)
- Functional status (e.g., ECOG performance status)
- Prior treatment history and timing constraints
- Laboratory and organ function thresholds
- Informed consent and regulatory requirements

Exclusion Criteria:

- Alternative diagnoses or disease subtypes
- Contraindicated comorbidities
- Active infections or autoimmune conditions
- Prior or concomitant therapies not permitted
- Safety-related laboratory abnormalities
- Other conditions posing excessive risk

Treatment Description

- Investigational product(s)
  - Dose, route of administration, schedule
- Control treatment(s)
  - Dose, route of administration, schedule
- Treatment duration and discontinuation rules

Study Endpoints

- Primary endpoint definition
- Secondary endpoint definitions
- Safety endpoints
- Timing of assessments

## **B.0.2 Clinical Diary Template**

Listing B.2: Abstract template for clinical diaries in research

Patient Identification

- Age, sex
- Date of consultation

Diagnosis and Disease Characteristics

- Primary diagnosis
- Disease stage or classification
- Molecular or biomarker status (if applicable)
- Relevant comorbidities or prior malignancies

Clinical History

- History of present illness (HPI)
  - Symptom onset, duration, severity
  - Diagnostic imaging or lab results
  - Previous interventions (surgical, pharmacological)
- Past medical history
- Surgical history
- Medication history

- Family medical history

#### Assessment at Consultation

- ECOG Performance Status or equivalent functional assessment
- Vital signs (weight, blood pressure, etc.)
- Relevant laboratory values or biomarkers
- Physical exam findings
- Imaging studies and results

#### Planned Interventions or Treatment

- Proposed pharmacological therapy
  - Drug names, doses, schedules
- Proposed surgical or procedural interventions
- Radiotherapy plans (if applicable)
- Monitoring or follow-up procedures

#### Treatment Rationale

- Explanation for treatment choice based on:
  - Disease stage and biology
  - Patient functional status
  - Safety considerations
  - Evidence from guidelines or prior studies

#### Follow-up and Outcomes Tracking

- Planned follow-up dates and assessments
- Metrics to track disease progression (e.g., tumor size, metastasis,
- Adverse events and safety monitoring
- Patient-reported outcomes (optional)

#### Comments / Notes

- Additional considerations or observations by clinician
- Any relevant patient preferences or social considerations

# Appendix C

## Background Knowledge

### C.1 Symbolic and Connectionist AI

This section is the study and interpretation of the document [Douglas, 2023].

Symbolic AI systems are explicitly designed and programmed, that is, human experts encode their domain knowledge directly into the system. The model receives a set of rules and an input, applying those rules to generate an appropriate output.

For example, in a grammar-based system:

Rule set:

Sentence  $\rightarrow$  Subject + Predicate

Subject  $\rightarrow$  Article + Noun

Article  $\rightarrow$  "the" | "a"

Predicate  $\rightarrow$  Verb + Object

Input: "The cat chases the mouse"

Application of rules:

"The"  $\rightarrow$  Article

"cat"  $\rightarrow$  Noun  $\rightarrow$  together form the Subject

"chases"  $\rightarrow$  Verb

"the"  $\rightarrow$  Article

"mouse"  $\rightarrow$  Noun  $\rightarrow$  together form the Predicate

Thus, the system determines that the sentence is grammatically correct.

However, this rule-based approach relies on fixed and absolute rules, which makes it difficult to apply to real-world situations where such absolutes rarely exist. Moreover, as problem complexity increases, the number of required rules grows exponentially, making maintenance and scalability impractical.

To overcome these limitations, probabilistic reasoning was introduced, along with

the development of extensive rule sets designed by experts, giving rise to Expert Systems.

In contrast, the Connectionist approach is inspired by the structure and functioning of the human brain, particularly the interconnected network of neurons, hence the term connectionist. This paradigm relies on statistics and information theory rather than explicitly defined rules. Connectionism represents the foundation of Machine Learning (ML) and, consequently, of Large Language Models (LLMs). In modern ML, a task is framed as a statistical inference problem based on a large dataset.

There are three main learning paradigms:

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning, in which the task is formulated as an objective function to be optimized.

According to Occam's Razor, simpler models tend to generalize better, [Soklakov, 2002]. Therefore, a model should include only the minimum number of parameters necessary to fit the dataset effectively.

Terms such as Deep Learning and Connectionism refer to the use of Neural Networks as the underlying computational structure.

Nonetheless, the Interpretability Problem remains one of the main challenges in deep learning. The large number of parameters in neural networks makes it extremely difficult to understand how the model reaches its conclusions or why a given input leads to a particular output.

## C.2 Language Models

Large Language Models (LLMs) are statistical models designed to encode a probability distribution over sequences of words. Formally, the probability of a sentence consisting of  $c$  words can be represented as:

$$P(W_1, W_2, \dots, W_c)$$

where  $W_1$  denotes the first word in the sequence.

### C.2.1 From Simple Models to Probabilistic Language Understanding

The simplest case, a 1-gram (unigram) model, assumes that each word is independent of all others. In such a case, the probability of the entire sentence is simply

the product of the individual word probabilities:

$$P(W_1, W_2, \dots, W_L) = \prod_{i=1}^L P(W_i)$$

where

$$P(W) = \frac{\text{Number of occurrences of } W \text{ in the corpus}}{\text{Total number of words in the corpus}}$$

Here, the corpus refers to a large collection of text used for estimating these probabilities.

However, modern LLMs are generative models, which means they can sample from this learned probability distribution to generate new sequences. Given a context, i.e., a sequence of words, the model estimates the probability of the most likely next word:

$$P(\text{next word}|\text{context})$$

## C.2.2 Examples of Prediction Tasks

- Masked Word Prediction Example:  
 $P(\text{went}|\text{The cat [BLANK] outside}) = 0.5$   
 $P(\text{sat}|\text{The cat [BLANK] outside}) = 0.2$   
 $P(\text{is}|\text{The cat [BLANK] outside}) = 0.1$   
 Each probability contributes to a total of 1 (100%). The input is “The cat ... outside” and the output is the missing token.
- Next Word Prediction  
 $P(\text{outside}|\text{The cat went}) = 0.7$   
 $P(\text{home}|\text{The cat went}) = 0.2$   
 $P(\text{to}|\text{The cat went}) = 0.1$

By sampling from this distribution and appending the predicted word to the existing sequence, the model can generate text one token at a time:

$$P(W_{n+1}|W_1, W_2, \dots, W_n)$$

Example:

$$\begin{aligned} P(W|\text{the cat}) &\rightarrow W = \text{went} \\ P(W|\text{the cat went}) &\rightarrow W = \text{outside} \\ P(W|\text{the cat went outside}) &\rightarrow W = \text{and} \end{aligned}$$

This iterative use of the model’s own output as the next input defines an autoregressive model, represented by:

$$\begin{aligned} P(\text{the cat went outside}) &= P(\text{the}) \\ &\quad \times P(\text{cat}|\text{the}) \\ &\quad \times P(\text{went}|\text{the cat}) \\ &\quad \times P(\text{outside}|\text{the cat went}) \end{aligned}$$

### C.2.3 Evaluating Language Models

To assess how well a language model approximates the true probability distribution of a corpus, the standard metric used is cross-entropy loss:

$$L = -\frac{1}{N} \sum_{i=1}^{N-n} \log P(W_{i+n} | W_1, W_{i+1}, \dots, W_{i+n-1})$$

The summation captures the surprise of each word in the training corpus, except for the first  $n$  words that serve as context. The term  $\frac{1}{N}$  averages the total surprise across the dataset. The logarithm penalizes confident but incorrect predictions: for example, if  $P = 0.01$ , then  $\log(P) = -4.6$ ; if  $P = 0.99$ , then  $\log(P) = -0.01$ . The negative sign transforms this measure into a loss function to be minimized during training.

$$P(W_{i+n} | W_1, W_{i+1}, \dots, W_{i+n-1})$$

represents the probability the model assigns to the next word. For example, if  $i = 1$  and  $n = 3$ , for the sentence “The cat hunts the mouse”, we would be computing  $P(\text{the} | \text{The cat hunts})$ .

### C.2.4 Training Process

The training process of a language model can be summarized as follows:

1. A sequence of words is provided to the model, which predicts the next word and assigns probabilities to all vocabulary tokens.
2. The loss  $L$  is calculated to measure how far the prediction is from the true target.
3. The error is propagated backward through the network (backpropagation).
4. Model parameters are updated via gradient descent.
5. The process repeats iteratively until convergence.

## C.3 Simpler Generative Language Models (N-gram)

Through the statistical analysis of neighbouring words, it is possible to capture quantitative properties, and even aspects of meaning, of words. The simplest expression of this idea is the co-occurrence matrix.

Language models operate on tokens rather than whole words. For example:

- “Supersymmetrization”  $\rightarrow$  “Super”, “Symmetry”, “ization”

This process, called tokenization, helps models to better understand the structure and meaning of language.

### C.3.1 Co-occurrence Matrix

Let  $w$  be the set of words in a corpus, and let  $|w|$  denote its cardinality. We define the N-gram co-occurrence matrix  $M_N$ , which is of size  $|w| \times |w|$ . Each row and column represents a word, and each entry  $(w, w')$  counts the number of N-grams in the corpus that contain both words.

Example corpus: “O gato caça o rato. O cão guarda a casa”

To create a 3-gram co-occurrence matrix, we extract trigrams from the corpus. For instance, checking for “o” and “gato”:

- “O gato caça”  $\rightarrow$  contains both
- “gato caça o”  $\rightarrow$  contains both
- “caça o rato”  $\rightarrow$  does not contain both

The resulting matrix is represented below:

|        | o | gato | caça | rato | cão | guarda | a | casa |
|--------|---|------|------|------|-----|--------|---|------|
| o      | - | 2    | 2    | 3    | 2   | 1      | 0 | 0    |
| gato   | 2 | -    | 2    | 0    | 0   | 0      | 0 | 0    |
| caça   | 2 | 2    | -    | 1    | 0   | 0      | 0 | 0    |
| rato   | 3 | 0    | 1    | -    | 1   | 0      | 0 | 0    |
| cão    | 2 | 0    | 0    | 1    | -   | 2      | 1 | 0    |
| guarda | 1 | 0    | 0    | 0    | 2   | -      | 2 | 1    |
| a      | 0 | 0    | 0    | 0    | 1   | 2      | - | 1    |
| casa   | 0 | 0    | 0    | 0    | 0   | 1      | 1 | -    |

Table C.1: 3-gram Co-occurrence Matrix

Once this matrix is built, word embeddings can be derived simply by taking the corresponding column (or row) vector of the word:

$$\text{gato} \Rightarrow [2, 0, 2, 0, 0, 0, 0, 0, 0]$$

This vector provides a distributional representation of the word.

However, such vectors tend to have very high dimensionality. To address this, dimensionality reduction techniques such as Principal Component Analysis (PCA) are applied. The resulting lower-dimensional vectors are then fed into the model.

### C.3.2 Predicting the Next Word

To predict the most probable next word, the model outputs a vector  $v$ , and the system selects the word  $w$  that maximizes the inner product between  $v$  and its embedding:

$$v \cdot \iota(w)$$

where  $\iota(w)$  is the embedding of word  $w$  derived from the co-occurrence matrix.

The vector  $v$  itself is generated by a function  $F$ , which may be implemented as a Neural Network, Transformer, or other architectures.

To convert the similarity scores into a probability distribution, the Boltzmann distribution is applied:

$$P(w) = \frac{e^{\beta v \cdot \iota(w)}}{\sum_{w'} e^{\beta v \cdot \iota(w')}}.$$

- $v \cdot \iota(w)$ : measures similarity between the word and the predicted vector.
- $e^{\beta v \cdot \iota(w)}$ : transforms scores into positive values and amplifies differences.
- $\sum_{w'} e^{\beta v \cdot \iota(w')}$ : ensures probabilities sum to 1.

### C.3.3 Semantic Properties of Word Embeddings

It can be observed that word embeddings capture semantic relationships. For example:

$$\iota(\text{king}) - \iota(\text{man}) + \iota(\text{woman}) \approx \iota(\text{queen})$$

This phenomenon arises from the statistical regularities of word co-occurrences, and requires a large corpus to emerge clearly.

### C.3.4 Generalized Autoregressive L-gram Model

From these ideas, we can define a general autoregressive L-gram model:

1. Map the  $L$  input words  $w_i$  to their embedding vectors  $\iota(w_i)$ .
2. Apply the function  $F$  to this list of vectors to produce a prediction vector  $v$ .
3. Use the Boltzmann distribution to obtain a probability distribution over all possible words.

Regarding the function  $F$ :

- A simple Feedforward Network can be used, but such networks expect a single input, whereas we have  $L$  vectors, one per word.
- One possible solution is to concatenate these vectors into a single input.

However, this approach proved limited, as transformer architectures later emerged and surpassed it. A major challenge is that predicting the next word requires remembering previous context.

A simple improvement involves adding memory, which leads to Recurrent Neural Networks (RNNs). RNNs maintain a state vector  $s_i$  that updates with each word:

$$(v_{i+1}, s_{i+1}) = F(s_i, v_i, v_{i-1}, v_{i-2}, \dots, v_{i-k+1})$$

where:

- $(v_{i+1}, s_{i+1})$  is the output of  $F$ ;  $v_{i+1}$  predicts the next word, and  $s_{i+1}$  is the updated memory state.
- $F(s_i, v_i, v_{i-1}, \dots, v_{i-k+1})$  is the input to  $F$ ;  $s_i$  encodes memory so far, while  $v_i, v_{i-1}, \dots$  provide immediate context within a window of  $k$  words.

## C.4 Recipe for a Large Language Model

A Transformer can be defined as a function  $F$  that takes a list of vectors as input and produces an output vector. This represents a generalization of Feedforward Networks (FFNs), with an important distinction: Transformers account for permutation symmetry, meaning they can handle sequences where the order of inputs matters.

A Transformer is composed of two key types of layers:

### C.4.1 Feedforward Layers

A Feedforward Neural Network (FFN) is applied independently to each embedding vector. For each input vector  $u_i$ , the model applies a learned transformation to produce an output vector:

$$v_i = F_{\text{FFN}}(u_i)$$

### C.4.2 Attention Mechanism

The Attention mechanism allows each output vector to attend to, or focus on, other input vectors in the sequence. This enables the model to determine which parts of the input are most relevant for each word at a given moment.

The output vector  $v_i$  is computed as a weighted sum of all input vectors  $u_j$ , each multiplied by a learnable linear transformation matrix  $w$ :

$$v_i = w \sum_{j=1}^i c_{i,j} u_j$$

Where:

- $v_i$  is the output vector.

- $w$  is the transformation matrix.
- $c_{i,j}$  are the attention weights.
- $u_j$  are the input vectors.

$$c_{i,j} \equiv \frac{e^{u_i \cdot B \cdot u_j}}{\sum_{j=1}^i e^{u_i \cdot B \cdot u_j}}$$

The attention weights  $c_{i,j}$  determine how much attention the vector at position  $i$  gives to the vector at position  $j$ . These weights are computed using a softmax function, which converts numerical scores into normalized probabilities, ensuring that the sum of all  $c_{i,j}$  for a given  $i$  equals 1.

The matrix  $B$  is a learned parameter during training. It allows the model to learn which relational aspects between vectors are most relevant.

- When  $B = I$ , the dot product measures only similarity.
- When  $B$  is learnable, it can capture more complex relationships, such as those between verbs and subjects, even when they are not semantically similar.

### C.4.3 Attention Heads

Transformers use multiple attention heads operating in parallel rather than relying on a single mechanism. Each head has its own learnable matrices  $B$  and  $W$ , and produces output vectors of smaller dimension  $q$  (compared to the original embedding dimension  $p$ ).

If there are  $H$  heads, their outputs are concatenated, resulting in an overall embedding dimension:

$$p = H \times q$$

Analogy: Each attention head may focus on a different aspect of the input:

- Head 1  $\rightarrow$  object colour
- Head 2  $\rightarrow$  shape
- Head 3  $\rightarrow$  spatial relationships
- Head 4  $\rightarrow$  texture

This multi-head attention enables the model to integrate diverse types of information simultaneously.

## C.4.4 Positional Encodings

A pure attention mechanism is invariant to input order, meaning it cannot inherently distinguish between permutations of the same words. To address this, Transformers introduce Positional Encodings, numerical representations that encode the position of each word in the sequence.

These encodings are generated using trigonometric functions:

$$(e_{2i-1}, e_{2i}) = \left( \cos \frac{\text{position}}{10000^{\frac{2i}{d_{\text{pos}}}}}, \sin \frac{\text{position}}{10000^{\frac{2i}{d_{\text{pos}}}}} \right), \quad i \in \left\{ 1, \dots, \frac{d_{\text{pos}}}{2} \right\}$$

Where:

- $2i$  is the index within the vector.
- $d_{\text{pos}}$  is the dimensionality of the positional embedding (equal to the word embedding dimension, e.g.,  $d = 512$ ).
- $\frac{d_{\text{pos}}}{2}$  ensures the sine and cosine functions alternate along half the embedding dimension.

This mechanism provides positional context, ensuring that word order affects meaning, for instance: "The cat hunts" is not equal to "Hunts the cat". Although both sentences contain the same words, their meanings differ due to positional encoding. Each word's embedding is summed with its positional vector:

- Word embedding:  $[0.3, 0.9, 0.1, 0.4]$
- Positional encoding:  $[0.84, 0.54, 0.01, 1.00]$
- Combined input:  $[1.14, 1.44, 0.11, 1.40]$

Thus, even repeated words at different positions in the text will receive unique positional embeddings, enabling the model to preserve sentence structure and meaning.

## C.5 Attention Is All You Need

### C.5.1 Model Architecture

The Transformer architecture is composed of two main components: an encoder and a decoder. The encoder maps a sequence of symbolic representations  $(x_1, \dots, x_n)$  (e.g., words) into a sequence of continuous representations:

$$Z = (Z_1, \dots, Z_n)$$

The decoder then generates a sequence of output symbols  $(y_1, \dots, y_m)$  one element at a time, consuming previously generated symbols as additional input, making the model autoregressive.

### C.5.2 Encoder

The encoder consists of a stack of  $N = 6$  layers, each containing two sub-layers:

1. A multi-head self-attention mechanism
2. A feedforward neural network (FFN)

A layer normalization step is applied between sub-layers to stabilize and accelerate training.

### C.5.3 Decoder

The decoder also consists of  $N = 6$  layers. In addition to the two sub-layers found in the encoder, the decoder introduces a third sub-layer that performs multi-head attention over the encoder's output. Layer normalization is again applied between all sub-layers.

A key component in the decoder is the Masked Multi-Head Attention mechanism, which prevents each position from attending to subsequent (future) words. This ensures that predictions for position  $i$  depend only on positions  $< i$ .

Example:

- Input (to the decoder): [ $< s >$ , The, cat, chases, the]
- Target sequence: [The, cat, chases, the, mouse]

| ID | Decoder Input    | Target Output |
|----|------------------|---------------|
| 1  | [ $< s >$ ]      | The           |
| 2  | [ $< s >$ , The] | cat           |

Table C.2: Decoder Time Steps and Target Outputs

This design ensures that each word at position  $i$  can only see previous words, maintaining the causal structure required for autoregressive generation.

### C.5.4 The Attention Mechanism

The Attention mechanism maps a query and a set of key-value pairs to an output, computed as a weighted sum of the values. Each weight represents how compatible the query is with the corresponding key.

Example context:

- Sentence: "The cat chases"

- Query: “chases”
- Keys: [“The”, “cat”, “chases”]

Mathematically:

$$\begin{aligned}
 Q &= W_q \times \text{embedding}(\text{chases}) \\
 K &= W_k \times [\text{embedding}(\text{The}), \text{embedding}(\text{cat}), \text{embedding}(\text{chases})] \\
 V &= W_v \times [\text{embedding}(\text{The}), \text{embedding}(\text{cat}), \text{embedding}(\text{chases})]
 \end{aligned}$$

This allows the model to build contextualized and dynamic representations, capturing long-range dependencies and context-dependent meanings. Because the matrices  $W_q$ ,  $W_k$ , and  $W_v$  are learnable parameters initialized randomly, the model can learn diverse linguistic relations in parallel during training.

### C.5.5 Scaled Dot-Product Attention

The Scaled Dot-Product Attention function is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax} \left( \frac{QK^T}{\sqrt{d_k}} \right) V$$

where  $Q$  and  $K$  are matrices with dimension  $d_k$ .

- $QK^T$ : computes similarity scores between each query and key (compatibility matrix).
- $\text{softmax}(\cdot)$ : converts these scores into normalized probabilities.

This function processes multiple queries simultaneously, enabling efficient parallel computation.

### C.5.6 Applications of Attention in the Transformer

There are three types of attention mechanisms in the Transformer model:

#### 1. Encoder–Decoder Attention

- Queries ( $Q$ ): come from the decoder (previous decoder output).
- Keys ( $K$ ) and Values ( $V$ ): come from the encoder’s final output.
- Example: To generate “mouse”,  $Q$  = chases (Decoder) and  $K, V$  = [The, cat, chases, the, mouse] (Encoder).

#### 2. Self-Attention in the Encoder

- $Q, K, V$  all come from the same source, the encoder’s previous layer output.

- Each word can attend to all words in the input.
3. Masked Self-Attention in the Decoder
- $Q, K, V$  all come from the decoder.
  - Future words are masked to prevent information leakage.
  - Example: For “chases”, the model can only attend to [The, cat, chases].

## C.6 Prompting Strategies

Prompting is the practice of guiding a pretrained language model at inference time using instructions or examples, allowing it to perform new tasks without updating its internal parameters, [Hooper]. It has become the standard method for adapting LLMs to specific applications.

Prompt engineering techniques fall into three main categories:

- Manual prompting, zero-shot, pure instruction, one-shot, few-shot templates, where examples are embedded into the prompt to demonstrate the desired behavior, [Chen et al., 2025].
- Automatic prompt optimization, these methods search or optimize prompts to improve performance, techniques include discrete search over prompt variants, continuous optimization where prompts are learned as vectors, or LLMs generate/refine prompts, [Zhou et al., 2023].
- hybrid approaches, combine manual intuition with automatic tools. Human-in-the-loop prompt refinement allows iterative improvement based on model outputs, [Zhou et al., 2023].

These methods exist because LLM behavior is highly sensitive to prompt design.

A key phenomenon underlying prompting is In-Context-Learning (ICL), this happens when a LLM is given examples of input–output pairs in a prompt, it can perform a new task without any parameter updates. This capacity distinguishes modern LLMs from traditional classifiers and enables task adaptation purely via prompt context, [Zhou et al., 2024].

Some works try to explain these behavior through: Implicit Gradient-like Behavior, where the Transformer architecture may emulate gradientbased learning on the provided examples, Bayesian Inference Framework, where the model implicitly infers latent task variables conditioned on the examples within the prompt, and Structure Induction and Pattern Matching, where the model may leverage learned structural regularities in language data, recombining compositional patterns to generalize from prompt examples to new inputs, [Hahn and Goyal, 2023].

More advance prompting strategies exist, such as:

- Chain-of-Thought prompting, this type of prompts guide models to generate intermediate reasoning steps before producing a final answer [Wei et al., 2023].
- Self-correction and Task decomposition, Techniques like iterative self-critique or problem decomposition encourage the model to review and refine its outputs or to break down complex tasks into simpler sub-tasks, [Yang et al., 2024] and [Liu et al., 2025].
- Retrieval-augmented prompting, [Son et al., 2025].

Despite its effectiveness, prompting faces limitations: sensitivity to wording, context-length constraints, hallucinations and bias, vulnerability to adversarial prompts, and weaker robustness compared to fine-tuned models, [Hooper].

Prompt engineering matters because it enables rapid task adaptation, reveals emergent capabilities like ICL, and provides insight into how LLMs reason, generalize, and use their pretrained knowledge

Prompting allows models to perform a wide range of tasks without task-specific training, enabling access to AI systems for users without deep technical knowledge.

## C.7 Information Extraction

Information Extraction (IE), aims to convert unstructured text into structured representations such as entities, relations, and attributes. Traditionally, IE relied on rule-based systems and later statistical models like Hidden Markov Models, [Su], Maximum Entropy classifiers, [Osborne, 2002], and Conditional Random Fields, [Ding et al.], which required extensive feature engineering and linguistic preprocessing, .

Large Language Models fundamentally change this landscape. LLMs enable end-to-end information extraction directly from raw text, reducing or eliminating the need for task-specific architectures and extensive preprocessing pipelines, [Xu et al., 2024]. Through prompting and in-context learning, LLMs can perform entity and relation extraction in zero-shot or few-shot settings, adapting to new schemas without retraining, [Wei et al., 2022]. This capability has been demonstrated across multiple IE tasks, including named entity recognition, relation extraction, and event extraction, often achieving competitive or superior performance compared to traditional supervised models, .

Moreover, LLMs naturally integrate semantic knowledge acquired during pre-training, allowing them to leverage world knowledge and long-range context, which is particularly beneficial for relation extraction and complex document understanding.

However, LLM based IE still faces issues such as prompt sensitivity, hallucinations, and limited control over output structure. As a result, hybrid approaches

combining LLMs with symbolic constraints or post-processing are increasingly explored to improve reliability, [Xu et al., 2024].

# **Appendix D**

## **Mockups**

## TrialPilot

- Clinical Trials
- Clinical Diaries
- Patients

### Clinical Diary

Parameter extraction for the clinical diary - Clinical\_Diary-1254.txt

Clinical\_Diary-1254.txt

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Donec pharetra molestie lacus, ut viverra odio tincidunt in. Integer ultricies lobortis nulla in mattis. Vivamus ultricies ex vel mollis malesuada. Proin a dictum neque. Donec mauris leo, cursus a turpis non, venenatis rhoncus lorem. Duis pellentesque porttitor cursus. Nam at dignissim augue. Fusce justo eros, tincidunt et turpis quis, dictum facilisis eros. Fusce et ex placerat, tempus purus sed, porta erat. Nunc fermentum libero eget orci feugiat rhoncus. Nam volutpat ipsum at felis ullamcorper vulputate. Integer euismod auctor tristique. Nam sed ligula nec lacus placerat ultrices. Quisque vestibulum hendrerit neque a luctus. Nullam in augue sapien.

Nullam condimentum, nisi id convallis dapibus, urna est auctor arcu, et interdum tortor urna porttitor neque. Nunc quis metus porttitor, mattis tellus ac, accumsan felis. Pellentesque vestibulum felis id odio aliquam porta. Sed lacinia quam in fringilla venenatis. Duis vel libero ac sem malesuada euismod vitae eu tortor. Ut a elit sollicitudin, suscipit dolor in, tincidunt ante. Cras ut sapien sit amet arcu venenatis vulputate. Ut porta enim nisi. Fusce quis nulla id ante pharetra tincidunt et volutpat arcu. Nunc pulvinar sodales mauris, ut tristique mauris fringilla in. Morbi bibendum enim ut ante pellentesque, id suscipit metus malesuada. Integer vestibulum non massa eget rutrum. Suspendisse at augue eu dui tempus lobortis vel ac nisi. Nunc vestibulum lorem et risus luctus, eu interdum odio pellentesque.

Extracted parameters

|                                                    |                      |     |
|----------------------------------------------------|----------------------|-----|
| Name                                               | Date of Birth        | Age |
| Laertes Kleone                                     | 1968-04-23           | 57  |
| Diagnosis                                          | Staging              |     |
| Adenocarcinoma of the colon                        | Stage IIb (T3 N2 M0) |     |
| Molecular Status                                   |                      |     |
| KRAS wild-type, BRAF V600E negative, MSI-stable    |                      |     |
| Chemotherapy Protocol                              |                      |     |
| FOLFOX (Oxaliplatin + Leucovorin + 5-Fluorouracil) |                      |     |

Validate

Extract fields

Figure D.1: Clinical diary field extraction page

## TrialPilot

- Clinical Trials
- Clinical Diaries
- Patients

### Clinical Diary

Upload the clinical diary for parameter extraction

Drag and drop or [browse](#) your Clinical Diary

Clinical\_Diary-1254.txt  
5.7 MB of 7.8 MB  
Uploading... 0%

Figure D.2: Clinical diary upload page

